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Extracellular vesicle-mediated release of bis(monoacylglycerol)phosphate is regulated by LRRK2 and Glucocerebrosidase activity

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eLife Assessment

This **useful** study presents the potentially interesting idea that LRRK2 regulates cellular BMP levels and their release via extracellular vesicles, with GCase activity further modulating this process in mutant LRRK2-expressing cells. However, some of the evidence supporting these conclusions remains **incomplete**, and additional work is suggested under certain conditions. Overall, the study will be of interest to cell biologists working on Parkinson's disease.

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Abstract

The endolysosomal phospholipid, bis(monoacylglycerol)phosphate (BMP), is aberrantly elevated in the urine of Parkinson's patients with mutations in genes encoding leucine-rich repeat kinase 2 (LRRK2) and glucocerebrosidase (GCase). Because BMP resides on and regulates the biogenesis of endolysosomal intraluminal membranes that become extracellular vesicles (EVs) upon release, we hypothesized that elevated urinary BMP may be driven by increased exocytosis of BMP-enriched EVs. To test this hypothesis, we analyzed BMP metabolism and EV-associated release of BMP in wild type (WT) and R1441G LRRK2-expressing mouse embryonic fibroblast (MEF) cells. Using immunofluorescence microscopy and transmission electron microscopy we detected structural alterations in endolysosomes and antibody-accessible BMP pool, indicating that mutant LRRK2 affects endolysosomal homeostasis. Biochemical analyses of isolated EV fractions confirmed the effect on endolysosomes by showing an increase in LAMP2-positive EVs in mutant cells, which was partially restored by LRRK2 kinase inhibition but further augmented, albeit to a more variable extent, by GCase inhibition. Using mass spectrometry, we detected an overall increase in total di-22:6-BMP and total di-18:1-BMP in cell lysates from mutant LRRK2 MEFs compared to WT cells. Inhibition of LRRK2 kinase partially restored cellular BMP levels, whereas inhibition of GCase further increased the BMP content. In isolated EVs from LRRK2 mutant cells, LRRK2 inhibition decreased BMP content whereas GCase inhibition tended to increase it. Using metabolic labeling experiments, we demonstrated that the increase in cellular BMP content is not due to an increase

in BMP synthesis, even though we observed an increase in BMP synthesizing enzyme, CLN5, in LRRK2 mutant MEFs and patient-derived fibroblasts. Finally, pharmacological modulation of EV release and live total internal reflection fluorescence (TIRF) microscopy of endolysosomal exocytosis in human G2019S LRRK2 fibroblasts, further confirmed that BMP release is associated with EV secretion. Together, these results establish LRRK2 as a regulator of BMP levels in cells and its release through EVs and suggest that GCase activity further modulates this process in LRRK2 mutant cells. Mechanistic insights from these studies have implications for the potential use of BMP-positive EVs as a biomarker for Parkinson's disease and associated treatments.

Introduction

Variants in the gene encoding Leucine rich repeat kinase 2 (*LRRK2*) and β -glucocerebrosidase (GCase; *GBA1*) together constitute the largest contributors to familial and sporadic Parkinson's disease (PD)¹. Although G2019S is the most prevalent PD-associated *LRRK2* mutation, R1441G is more penetrant and is associated with an earlier disease onset^{2–4}. A shared mechanism of pathogenesis among all PD-causing *LRRK2* mutations is increased *LRRK2* kinase activity, which leads to phosphorylation of its cognate Rab GTPase substrates, including Rab8, Rab10 and Rab12, that are key regulators of intracellular vesicle trafficking⁵. A prevailing hypothesis of pathogenesis in *LRRK2* PD cases is that *LRRK2* phosphorylation of Rabs results in defective autophagy and endolysosomal homeostasis and defective ciliogenesis in the nigrostriatal circuit^{5,6}.

GBA1 encodes a lysosomal hydrolase, glucocerebrosidase (GCase), essential for glycosphingolipid degradation. Homozygous loss-of-function mutations in *GBA1* cause Gaucher's disease, a lysosomal storage disorder characterized by cytotoxic accumulation of lipid substrates of GCase, glucosylceramide and glucosylsphingosine; heterozygous carriers have a 5-10 fold increased risk of PD^{7,8}. The predominant hypothesis for the pathogenic mechanism of *GBA1* variants is related to aberrant aggregation and clearance of alpha-synuclein in a feed-forward cascade^{9,10}, a hypothesis being tested clinically using GCase activators. Additionally, we and others previously showed that GCase enzymatic function is modulated by *LRRK2* kinase activity^{11,12}. Altogether, it is possible that dysregulation of the endolysosomal biology is associated with pathogenic variants of both *LRRK2* and *GBA1*^{6,13}.

Bis(monoacylglycerol)phosphate (BMP, also known as LBPA or Lysobisphosphatidic acid) is an atypical, negatively-charged glycerophospholipid important for maintaining endolysosomal homeostasis¹⁴. BMP resides primarily in endolysosomal intraluminal vesicles (ILVs) where it participates in multiple processes, including cholesterol egress, exosome biogenesis and lipid catabolism^{14,15}. Of note, BMP acts together with Saposin-C as enzymatic cofactors of GCase^{16,17}. Multiple BMP isomers are present in cells, based on differences in their fatty acyl chains and their positions relative to the sn-2 and sn-3 carbon atoms on each of its two glycerol backbones^{14,15}; the sn-2:sn-2' isomer is the proposed active form^{18,19}. Dioleoyl-BMP (di-18:1-BMP) is the predominant form in numerous cell lines analyzed^{20–23} and didocosaheptaenoyl-BMP (di-22:6-BMP) is one of the most abundant species in the brain^{24,25}. Fatty acyl composition is thought to influence BMP biophysical properties and function²⁶.

We and others recently reported elevated levels of total di-18:1-BMP and total di-22:6-BMP species in the urine of carriers of PD-associated *LRRK2* and *GBA1* mutations^{27–29}. Although we did not observe a correlation between urinary BMP and PD progression²⁸, these studies underscore BMP's utility as a patient enrichment and target modulation biomarker in therapeutic trials³⁰. Based on our previous findings, in the present study we focused on these two BMP species.

Given the specific intracellular localization of BMP (in ILVs; exosomes when released) and its proposed roles in exosome biogenesis¹⁹, we hypothesized that its increased levels in PD biofluids may be a consequence of an increase in the secretion of BMP containing extracellular vesicle (EV). To gain mechanistic insight into the regulation and biological significance of BMP release, we monitored BMP release in EVs and performed kinetic analyses of BMP biosynthesis in mouse embryonic fibroblasts (MEFs) from wild-type (WT) or R1441G *LRRK2* knock-in mice. Moreover, endolysosomal exocytosis was visualized and quantified in skin fibroblasts derived from both

healthy donors and G2019S LRRK2 PD-patients. In addition, we examined the regulation of BMP by a LRRK2 kinase inhibitor, MLI-2, and a GCase inhibitor, conduritol β -epoxide (CBE), to gain insights into the role of BMP in potential therapeutic vs. pathophysiological responses, respectively. Our data indicate that LRRK2 kinase activity modulate BMP release in EVs and further suggest that GCase function may contribute to this process.

Results

Analysis of BMP-positive endolysosomes in R1441G LRRK2 mouse embryonic fibroblasts

We first performed immunofluorescence microscopy of BMP-positive endolysosomes in both WT and R1441G LRRK2 knock-in MEFs. The overall antibody-accessible BMP pool was significantly decreased in mutant LRRK2-expressing cells compared to WT (Figure 1A [B](#)). In contrast, endolysosomal LAMP2 fluorescence was higher in mutant LRRK2 MEFs (Figure 1A [C](#)). Previous studies found structural defects in endolysosomes of different cell types harboring pathogenic LRRK2 mutations^{31–33}. Since BMP is specifically enriched in intra-luminal vesicles (ILVs) in the endolysosomes where it regulates important catabolic reactions, we examined multivesicular endosome (MVE) morphology of these cells. In agreement with previous findings³³, transmission electron microscopy revealed decreased MVE area in R1441G LRRK2 MEFs (average area= 0.11 μm^2) compared to WT MEFs (average area= 0.27 μm^2) (Figure 1D [E](#)). In addition, the overall endolysosomal ILV number was diminished in the R1441G LRRK2 cell line (Figure 1F [G](#)). These data suggest that the pathogenic LRRK2 mutation alters BMP-positive endolysosome morphology and ILV content.

Investigating the impact of LRRK2 and GCase activities on extracellular vesicle release modulation

Given the previously described roles of BMP in ILV/exosome biogenesis, we investigated whether the alterations observed in antibody-accessible BMP and MVE ILV number in LRRK2 mutant MEFs could be explained by changes in EV release. We isolated and characterized EVs from both WT and R1441G MEFs and also assessed the effects of LRRK2 and GCase pharmacological inhibitors, MLI-2 and CBE, respectively on EV content and number. Inhibition of LRRK2 kinase and GCase enzymatic activities was confirmed, respectively, by monitoring phospho-Rab10 levels in whole cell lysates and performing a fluorescence-based GCase activity assay in cells (Figure 2A [B](#)). Consistent with our immunofluorescence data, upregulation of LAMP2 was observed in R1441G LRRK2 whole cell lysates (WCL; Figure 2 [A](#), C, D and E), as previously reported in LRRK2 G2019S knock-in mouse brain³⁴. MLI-2 treatment for 48 hours partially reversed this phenotype (Figure 2 [A](#), C, E), suggesting that aberrant LRRK2 kinase activity influences endolysosomal homeostasis.

We next analyzed the profiles of isolated EV fractions from both wild-type (WT) and R1441G LRRK2 mutant MEFs. Both LAMP2 and Flotillin-1 were assessed to explore alterations in release of EV subpopulations; LAMP2 is enriched in ILV-derived EVs while Flotillin-1 is also seen in plasma membrane-derived ectosomes that reflect outward budding of the plasma membrane.^{35–37} Biochemical analysis revealed an elevation of LAMP2, but not Flotillin-1, in EVs derived from R1441G LRRK2 MEFs compared to those from WT MEFs (Figure 2C [F](#), G). Upon MLI-2 treatment, a modest but significant decrease in both EV-associated protein markers was observed in isolated EV fractions from mutant LRRK2 MEFs (Figure 2C [F](#), G). In contrast, inhibition of GCase activity in mutant LRRK2 cells led to an increase in LAMP2, but not Flotillin-1, in isolated EVs compared to those derived from WT MEFs (Figure 2C [F](#), G). Finally, analysis of WT cell-derived EVs revealed no major differences in EV marker levels between untreated (control) and MLI-2- or CBE-treated conditions (Supplemental Figure 1A–C).

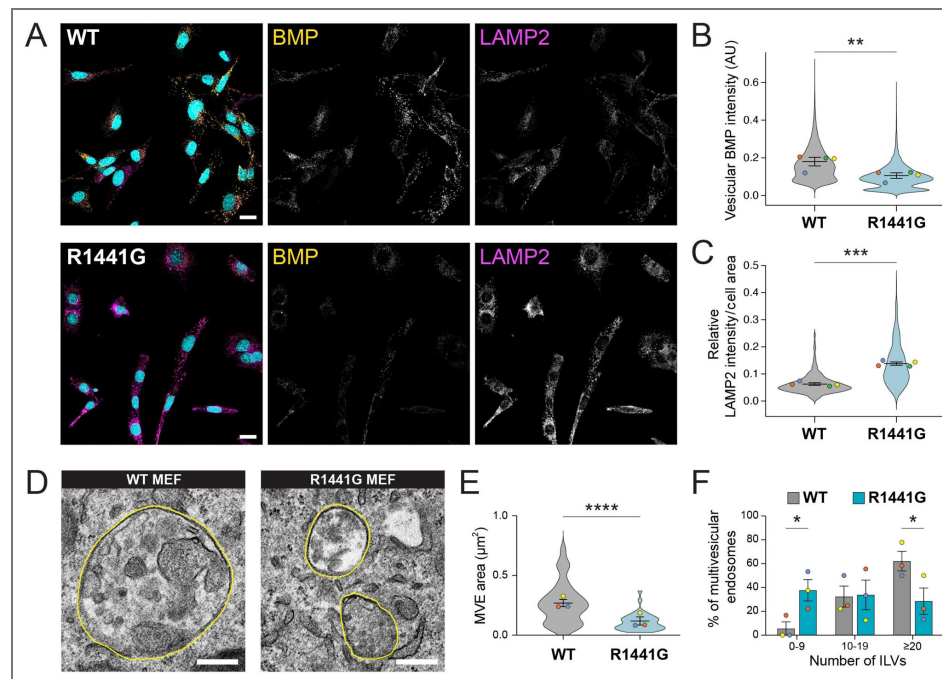


Figure 1. Alterations in antibody-accessible BMP and endolysosomal morphology in R1441G LRRK2 MEF cells.

(A) Confocal microscopy of endogenous BMP (green) and LAMP2 (red) immunofluorescence in WT and R1441G LRRK2 MEFs. Scale bar: 20 μm . **(B-C)** Quantification of vesicular BMP intensity **(B)** and LAMP2 relative intensity **(C)** per cell area. Colored dots represent mean value from 4 independent experiments and violin plots show the distribution of individual cell data. Significance determined by two-tailed paired t test $**p < 0.01$, $***p < 0.001$. **(D)** Representative transmission electron microscopy (TEM) images of Multivesicular endosomes (MVE) from WT and R1441G LRRK2 MEFs. MVB periphery highlighted in yellow. Scale bar: 250 nm. **(E)** MVE area (μm^2) quantification in WT and R1441G LRRK2 mutant cells. Colored dots represent mean values from 3 independent experiments and violin plots show the distribution of individual cell data (35-45 cells/group). **(F)** Quantification of Intraluminal Vesicles (ILVs) per MVE in WT and R1441G LRRK2 MEF cells. The number of ILVs per MVE are binned in three groups and plotted as a percentage of MVE from the total population of each experiment independently. Data from 3 independent experiments (mean $\pm$ SEM). Significance determined by two-tailed unpaired t test **(E)** and ordinary two-way ANOVA, uncorrected Fisher's LSD **(F)** $*p < 0.05$, $***p < 0.0001$.

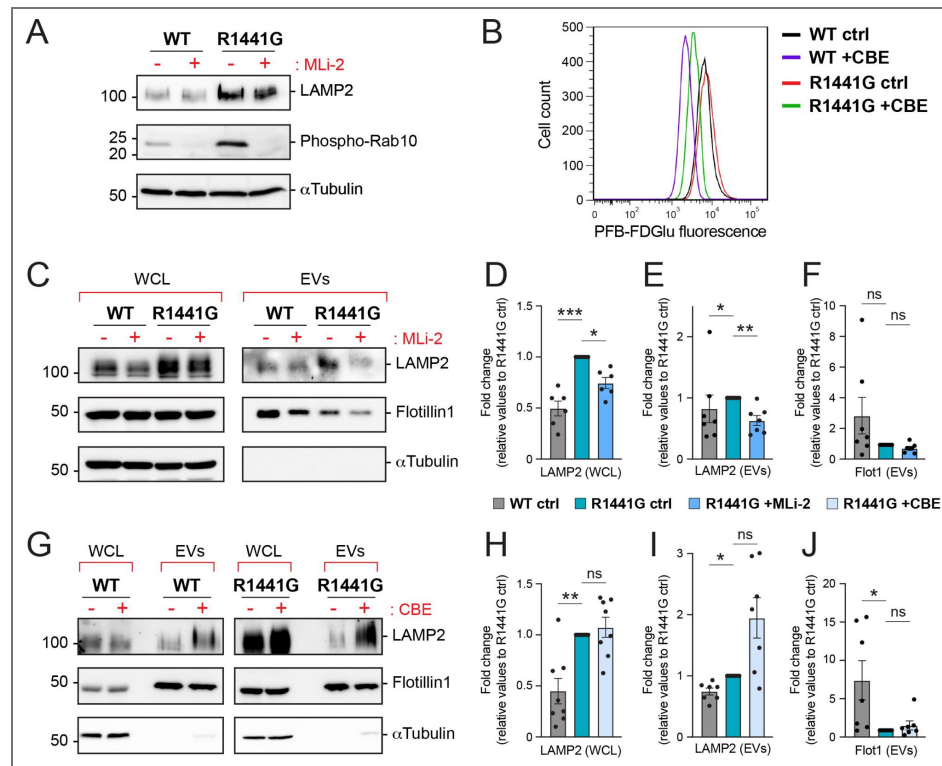


Figure 2. LRRK2 and GCase activities modulate extracellular vesicle production.

(A) Whole cell lysates from WT and R1441G LRRK2 MEF cells treated with 200nM MLI-2 for 24h were analyzed by immunoblotting. Representative images of LAMP2, phospho-Rab10 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa. (B) Flow cytometry measurement of GCase activity using PFB-FDGLu fluorescent GCase substrate in WT and R1441G LRRK2 mutant MEF cells treated with 300μM CBE for 24h. (C and G) Whole cell lysates (WCL) and isolated extracellular vesicles (EVs) from WT and R1441G LRRK2 mutant MEF cells treated with 200nM MLI-2 (C) or 300μM CBE (G) for 48h were analyzed by immunoblotting. Representative images of LAMP2, Flotillin-1 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa. Immunoblots for LAMP2 and Flotillin-1 in EV fractions required longer exposure times to visualize clear signals across all conditions. (D-F and H-J) Quantification of LAMP2 and Flotillin-1 levels relative to R1441G LRRK2 MEF cells in whole cell lysates (WCL). (D and H) and isolated EVs (E, F, I and J). Data from 6-8 independent experiments (mean ± SEM). Significance determined by Kruskal-Wallis test followed by an uncorrected Dunn's post hoc test compared to R1441G LRRK2 control *p<0.05, **p<0.01, ***p<0.0001; ns, not significant.

To complement these data, we conducted nanoparticle tracking analysis (NTA). Isolated EV fractions from WT and R1441G LRRK2 cells exhibited comparable particle size distributions (Supplemental Figure 1D). Treatments with MLI-2 and CBE yielded measurable quantitative changes in EV concentrations that, despite not reaching statistical significance, gave similar trends as those seen in our biochemical analyses (Supplemental Figure 1E), potentially reflecting the inherent variability of NTA due to its inability to distinguish EVs from non-vesicular particles^{38,39}. For the R1441G MEF cells, MLI-2 decreased EV concentration while CBE increased EV particles per ml, in agreement with the effects observed in our biochemical analysis.

Altogether, these results suggest that EV secretion is influenced by LRRK2 kinase and, to a more variable degree, by GCase hydrolase activities; while MLI-2 treatment decreases EV release, CBE appears to increase it.

Targeted BMP lipid analysis in intact cells and extracellular vesicles

Ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) was used to measure quantitatively the impact of LRRK2 and GCase activities on BMP isoform abundance in cells and isolated EV fractions. Remarkably, an overall increase in total BMP isoforms was detected in mutant LRRK2 MEF cell lysates (Figure 3A-D [↗](#)), with di-22:6-BMP and di-18:1-BMP as the major species. It is important to note that mass spectrometry-based methods detect total BMP content while antibody staining (Figure 1A [↗](#), B) only detects so-called “antibody-accessible” BMP⁴⁰. The mass spectrometry-based increase in total BMP in LRRK2 mutant cells indicates that this pool is less antibody-accessible than that present in wild type cells. Alternately, the anti-BMP antibody may be less specific and detect other analytes.

In R1441G cells, LRRK2 inhibition for 48 hours with MLI-2 decreased total di-22:6-BMP and total di-18:1-BMP levels by ~20% (Figure 3A and B [↗](#)). Conversely, inhibition of GCase increased total BMP levels, reaching significance for di-22:6-BMP levels compared with untreated, R1441G LRRK2 cells (Figure 3C [↗](#)). No statistically significant differences in intracellular BMP levels were observed in WT LRRK2 MEFs upon LRRK2 or GCase inhibition (Supplemental Figure 1F, G), suggesting a dominant role of mutant LRRK2 activity in BMP regulation. Overall, di-oleoyl (18:1)- and di-docosahexaenoyl (22:6)-BMP species were the most abundant in MEFs, whereas other isoforms, such as di-arachidonoyl (20:4)- and dilinoleoyl (18:2)-BMP, were present at lower levels but were also consistently elevated in mutant LRRK2 MEFs (Supplemental Figure 1H). These data suggest that there are LRRK2-independent clonal differences leading to differences in basal BMP content between WT and mutant MEF cells; however, such clonal variation does not impact the effect of MLI-2 or CBE treatment in R1441G cells.

Analysis of isolated EV fractions detected only the di-18:1-BMP isoform, with no evidence of di-22:6-BMP. Although a trend toward higher di-18:1-BMP levels was observed in EVs derived from WT cells compared to those from mutant LRRK2 cells, this difference was not statistically significant (Figure 3E [↗](#)). Treatment of LRRK2 R1441G MEFs with MLI-2 resulted in a significant partial decrease in EV-associated total BMP (Figure 3E [↗](#)), consistent with previous observations in non-human primates and PD mouse models that showed decreased extracellular urinary BMP upon LRRK2 kinase pharmacological inhibition^{41–43}. In contrast, inhibition of GCase activity yielded an opposite, albeit not significant, trend (Figure 3E [↗](#)). These latter findings may be explained by the observation that GCase inhibition by CBE was less pronounced in LRRK2 R1441G cells compared to WT cells under identical concentration and treatment duration conditions (Figure 2B [↗](#)). Finally, no significant differences in EV-associated BMP abundance between untreated (control) and MLI-2- or CBE-treated WT LRRK2 MEFs were observed (Supplemental Figure 1I).

In addition to analyzing BMP, we also examined GCase lipid substrates in both cells and isolated EVs. Targeted quantification of sphingolipid species revealed elevated GCase substrate abundance in both CBE-treated WT and mutant LRRK2 cells, validating the treatment paradigm used in this study. Interestingly, CBE-mediated accumulation of GCase substrates was more evident in R1441G

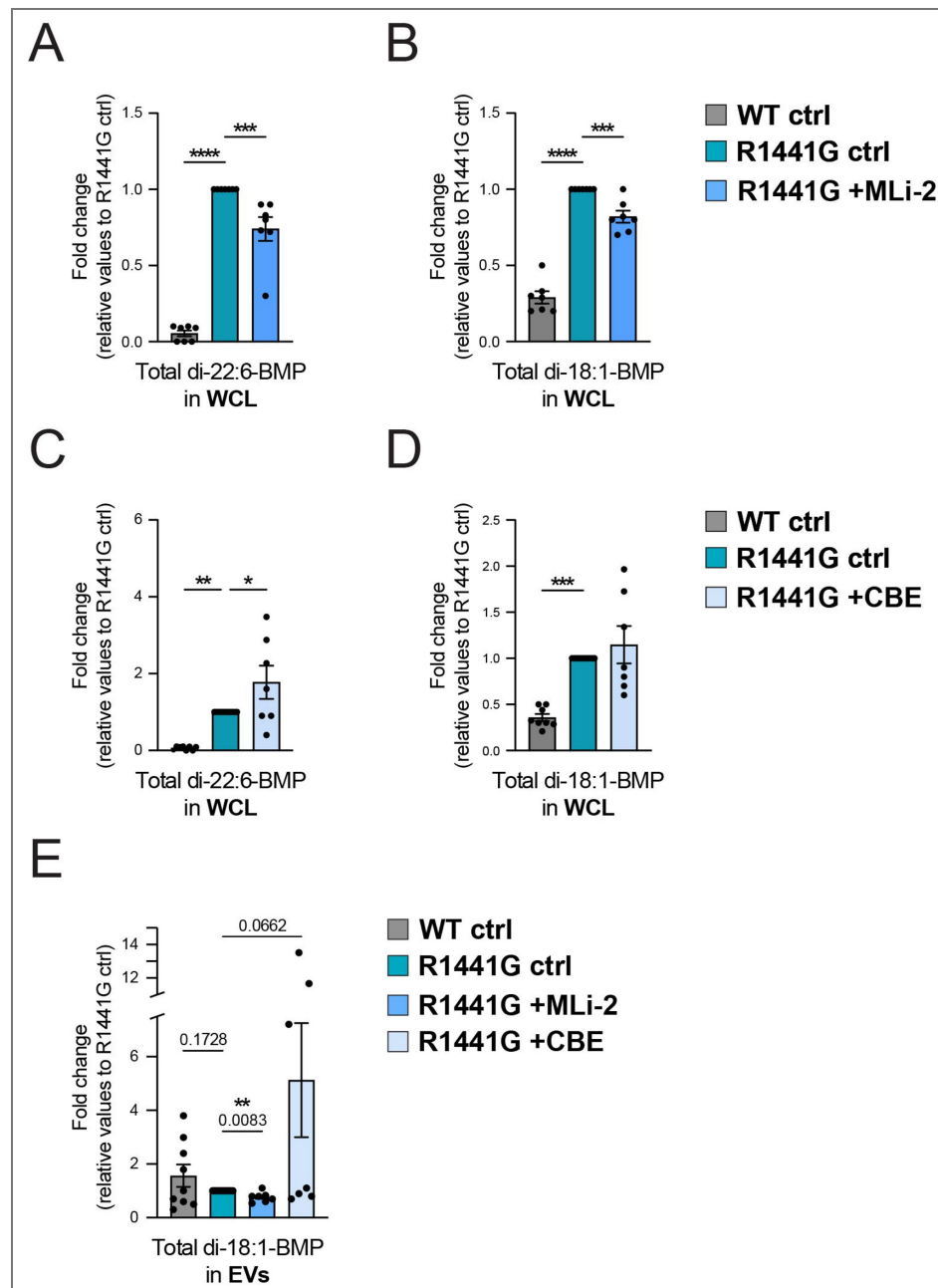


Figure 3. Targeted lipid pathway analysis of BMP abundance in cellular and isolated EV fractions.

(A-D) UPLC-MS/MS determination of BMP isoforms normalized to protein content from cells treated with 200nM MLI-2 (A-B) or 300μM CBE (C-D) for 48h. Data shown as fold change relative to untreated R1441G LRRK2 MEF cells. Only BMP isoforms that were detected are shown. (E) UPLC-MS/MS determination of BMP isoforms normalized to protein content in EVs isolated from cells treated with 200nM MLI-2 or 300μM CBE for 48h. Only BMP isoforms that were detected are shown. Data from 3-6 independent experiments (mean ± SEM). Significance determined by ordinary one-way ANOVA, uncorrected Fisher's LSD (A-D) and one-way ANOVA with the Geisser-Greenhouse correction, uncorrected Fisher's LSD (E) *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

LRRK2 than in WT cells (Supplemental Figure 2A). This suggests that, despite a less pronounced direct effect on GCase activity (Figure 2B), the R1441G mutation might contribute to broader endolysosome dysfunction, which could alter the dynamics of substrate accumulation. On the other hand, analysis of isolated EV fractions revealed lower levels of glucosylceramide, galactosylceramide, and glucosylsphingosine in EVs from R1441G LRRK2 cells compared to those from WT MEFs, but no significant differences were detected between control or CBE-treated conditions independent of LRRK2 mutation status (except for glucosylsphingosine, which showed an increase in WT-EVs upon GCase inhibition; Supplemental Figure 2B).

Altogether, given that BMP is specifically enriched in ILVs (which become exosomes upon release), the data presented above support our biochemical analysis (Figure 2C, E, G, I) and suggest that LRRK2 activity regulates BMP release in association with LAMP2-positive exosomes, whereas GCase activity appears to have a more variable effect under the tested conditions.

BMP biosynthesis is not influenced by LRRK2 kinase and GCase activities

To rule out the possibility that differences in cellular and EV-associated BMP levels following MLI-2 or CBE treatment of R1441G cells were caused by alterations in BMP metabolism rather than changes in membrane trafficking and EV release, we performed metabolic labeling experiments using heavy (H) isotope-labeled 22:6 and 18:1 fatty acids as BMP precursors.

Cells were pulsed for 25_min with heavy isotope BMP precursors and then washed and chased for different time points (Figure 4A). UPLC-MS/MS analysis allowed us to differentiate unlabeled versus semi-labeled (H,L'; only one heavy isotope-labeled fatty acid chain) or fully-labeled (H,H'; both fatty acid chains heavy isotope-labeled) BMP species (Figure 4A). Initial experiments were performed over 48h using mutant LRRK2 cells under control and LRRK2- or GCase-inhibition conditions; untreated WT cells were included for comparison. Strikingly, even at time 0 and throughout the length of the experiment, R1441G LRRK2 cells displayed substantially higher H,L'- and H,H'-22:6-BMP species than WT cells (left and middle plots in Figure 4B). H,L', but not H,H' 18:1-BMP was detected, which appeared to increase to a greater extent in R1441G LRRK2 cells at least during the initial 8h of the experiment (right plot, Figure 4B). Despite these differences between WT and mutant LRRK2 cells, neither inhibition of LRRK2 or GCase had a major impact in the kinetics of BMP synthesis or catabolism (Figure 4B).

In these experiments, semi-labeled BMP species gradually increased for up to 24 h. This likely reflects re-utilization of isotope-labeled lysophosphatidylglycerol (LPG) and fatty acid precursors generated after degradation of either H,L' or H,H' BMP by endolysosomal hydrolase PLA2G15⁴⁴. To better resolve initial BMP synthesis rates, we reduced the pulse labeling time and chase time to 60 min. In this set of experiments, only isotope-labeled 22:6-BMP species were detected (Figure 4C), possibly reflecting a preference for 22:6 BMP synthesis by MEFs. As before, even at time 0 and throughout the 60 min duration of the assay, 22:6-BMP levels were consistently higher in R1441G LRRK2 cells compared with WT MEFs. Again, no overall rate differences were seen between untreated and MLI-2- or CBE-treated cells. These experiments support the hypothesis that rather than altering BMP metabolic rates, LRRK2 and GCase activities influence EV-mediated BMP release.

Increased BMP synthase protein expression in mutant LRRK2 cells

To further investigate the biological significance of BMP upregulation observed in mutant LRRK2 MEF cells (Figure 3A-D; Figure 4B, C), we examined the expression of CLN5, a key lysosomal enzyme in the BMP biosynthetic pathway^{45,46}. Interestingly, biochemical analysis of total cell lysates revealed a significant fold-change increase in CLN5 protein levels in R1441G LRRK2 MEFs relative to WT cells (Figure 5A). Notably, 16 hour treatment with MLI-2 reduced CLN5 levels to a comparable fold-change extent (Figure 5B). To validate these findings in a human and disease-relevant context, we examined CLN5 protein expression in cultured fibroblasts derived from Parkinson's disease (PD) patients carrying the G2019S LRRK2 mutation. Consistent with our

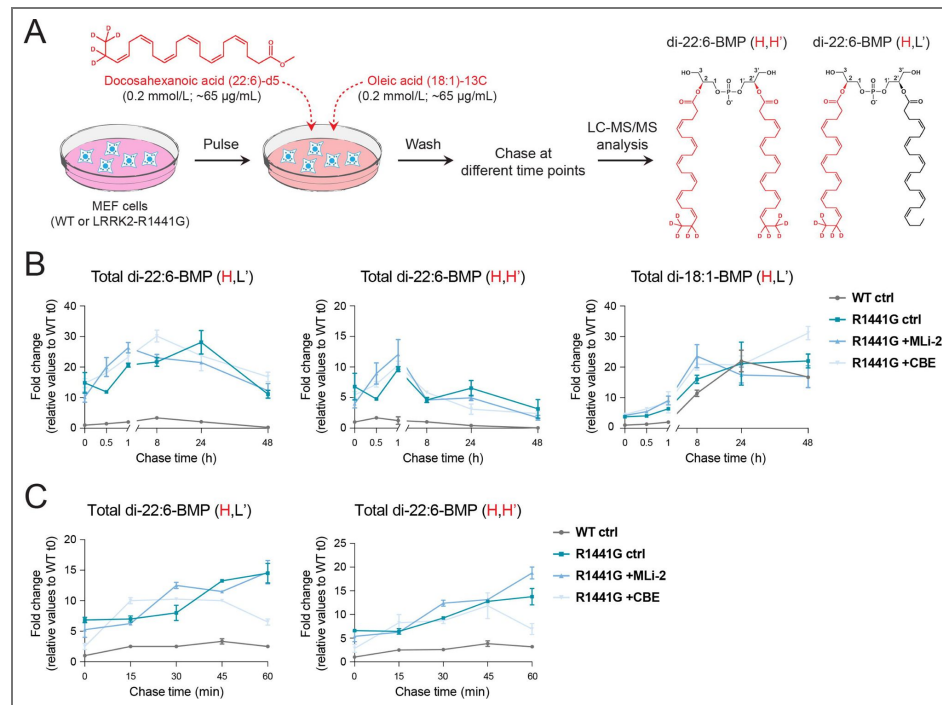


Figure 4. Inhibition of LRRK2 or GCase activities does not significantly impact BMP biosynthetic and catabolic rates.

(A) Schematic representation of the BMP metabolic labeling protocol with deuterated docosahexanoic acid and ^{13}C -labeled oleic acid. WT and R1441G mutant MEF cells were incubated with a pulse of DHA-d5 / OA- ^{13}C for 20min, followed by a chase for different times. Cells were then collected for subsequent BMP lipidomic analysis. Structures of unlabeled (L) and isotope-labeled (H) fatty acids are shown in black or red, respectively. **(B-D)** UPLC-MS/MS determination of BMP isoforms normalized to protein content from WT MEF cells and R1441G LRRK2 mutant MEF cells ± MLI-2 (200nM) or CBE (300µM). Long **(B-C)** and short **(D)** chase time points shown as fold change relative to WT control MEF cells time 0. Only BMP isoforms that were detected are shown. Data from 3 replicated experiments (mean ± SEM).

observations in MEF cells, CLN5 protein levels were reduced following 16 h MLI-2 treatment in a concentration-dependent manner (Figure 5C). Notably, the maximal reduction was observed at the same 200nM MLI-2 concentration used in MEF cell experiments, a dose that also did not induce observable cytotoxic effects in human fibroblasts. Similar results were obtained in patient-derived fibroblasts harboring the R1441G LRRK2 mutation (Figure 5D). Altogether, these data suggest that LRRK2 kinase activity may regulate CLN5 protein expression.

The upregulation of CLN5 may be due to an overall upregulation of lysosomal enzymes as LAMP2 levels were also increased (Figure 2A, C, E). Moreover, the observed baseline differences in BMP, as documented in the flux study above (Figure 4B and C), could result from CLN5 upregulation. The lack of significant changes in the BMP synthesis rate (Figure 4B and C) suggests either a limitation in substrate availability or that CLN5 is operating at maximal capacity.

BMP release is EV-mediated

Our data suggest that BMP is exocytosed in association with EVs and that LRRK2 and GCase activities modulate BMP secretion. To determine the magnitude of BMP release via EV secretion, we assessed the impact of pharmacological modulators of EV release. Treatment of WT MEFs with GW4869, a selective type 2-neutral sphingomyelinase inhibitor, decreased EV release as monitored by reduced levels of LAMP2 and Flotillin-1 in EV fractions (Figure 6A). Under these conditions, we observed a parallel decrease in exosomal BMP (Figure 6A, B). In contrast, enhancing EV release with bafilomycin A1 (B-A1), a pharmacological inhibitor of the endolysosomal proton pump V-ATPase that dramatically boosts EV exocytosis^{36,47}, resulted in the opposite trend (Figure 6A, C): EV markers and BMP levels increased. Biochemical analysis of EV release in R1441G LRRK2 cells showed similar behavior as in WT MEFs upon treatment with these agents (Figure 6A and Supplemental Figure 3). GW4869 inhibition of EV release increased cellular BMP content (Figure 6D), while enhanced EV release upon B-A1 treatment diminished intracellular total di-22:6-BMP and di-18:1-BMP (Figure 6E). Altogether, these results strongly support the notion that BMP is released by EV-mediated MVE exocytosis.

Monitoring in vivo endolysosomal exocytosis and EV release in patient-derived G2019S LRRK2 fibroblasts

All our previous experiments investigating the role of LRRK2 in EV release regulation were conducted using mouse-derived cells. To further validate these findings, we employed skin fibroblasts derived from healthy donors (control) and G2019S LRRK2 PD patients. We first performed an immunofluorescence and biochemical analysis of the BMP-positive endolysosomal compartment in three independent skin fibroblast cell lines from each cohort. Consistent with our observations in MEFs, we observed a decrease in BMP immunostaining in LAMP1-positive endolysosomes of G2019S LRRK2 cells compared to control (Figure 7A and B). LAMP1 levels remained unchanged between the two cohorts (Figure 7A and C).

Next, we biochemically characterized endolysosomes isolated by subcellular fractionation on discontinuous sucrose gradients. Interestingly, and consistent with previous studies reporting association of LRRK2 with isolated endolysosomes⁴⁸, a pool of LRRK2 was detected in fractions positive for CLN5 and the endolysosomal marker CD63, but negative for tubulin (Supplemental Figure 4A). As seen in our whole cell lysate analysis (Figure 5C), relative CLN5 levels appeared increased in G2019S LRRK2 fibroblasts (Supplemental Figure 4A, bottom fractions). However, the relative amount of CLN5 co-peaking with isolated CD63-positive endolysosomes was comparable to that in control cells (Supplemental Figure 4A, endolysosome fractions). These results suggests that, in these cells, a considerable pool of newly synthesized CLN5 may not have reached its final destination at the endolysosome and may instead be retained in the ER. Such a delay in CLN5 trafficking could explain why, despite the apparent increase in CLN5 protein levels also found in mutant LRRK2 MEFs, the BMP biosynthetic rate did not differ from that observed in WT MEFs. Despite these changes in CLN5 expression, no statistically significant differences were detected in the total protein levels of CD63 between control and G2019S LRRK2 human fibroblasts (Supplemental Figure 4B). Collectively, these data indicate that human cells carrying pathogenic

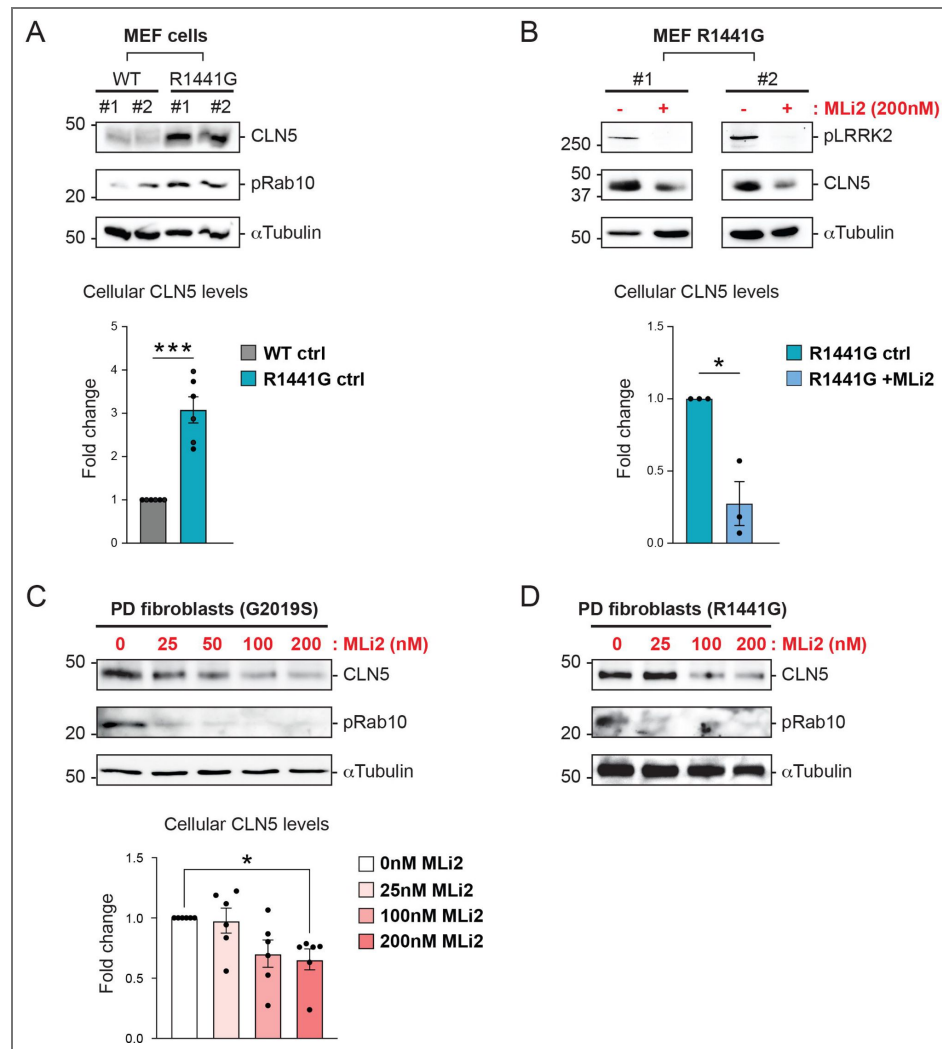


Figure 5. LRRK2 activity modulates CLN5 expression levels.

(A) Whole cell lysates from WT and R1441G LRRK2 MEF cells were analyzed by immunoblotting. Representative immunoblots of CLN5, phospho-Rab10 (pRab10) and α -Tubulin are shown from two (#1 and #2) out of six independent experiments. Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels relative to WT and R1441G LRRK2 MEF cells. **(B)** Whole cell lysates from R1441G LRRK2 MEF cells treated with 200nM MLI-2 for 24h were analyzed by immunoblotting. Representative immunoblots of phospho-LRRK2 (pLRRK2), CLN5 and α -Tubulin levels are shown from two (#1 and #2) out of three independent experiments. Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels in whole cell lysates of R1441G LRRK2 MEF cells untreated or MLI-2-treated. **(C)** Whole cell lysates from G2019S LRRK2 patient-derived fibroblasts treated with indicated increasing MLI-2 concentrations for 24h were analyzed by immunoblotting. Immunoblots of CLN5, phospho-Rab10 (pRab10) and α -Tubulin levels are shown from one representative experiment (n=6 per condition, obtained from two independent replicate experiments using fibroblast cell lines derived from three different patients). Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels relative to G2019S LRRK2 patient-derived fibroblasts treated with MLI-2 at the indicated concentrations. **(D)** Whole cell lysates from R1441G LRRK2 patient-derived fibroblasts treated with indicated increasing MLI-2 concentrations for 24h were analyzed by immunoblotting. Immunoblots of CLN5, phospho-Rab10 (pRab10) and α -Tubulin levels are shown from one representative experiment. Significance in **(A)** and **(B)** determined by two-tailed, unpaired *t* test; significance in **(C)** determined by Dunnett's One-way ANOVA test; **p*<0.05, ****p*<0.001.

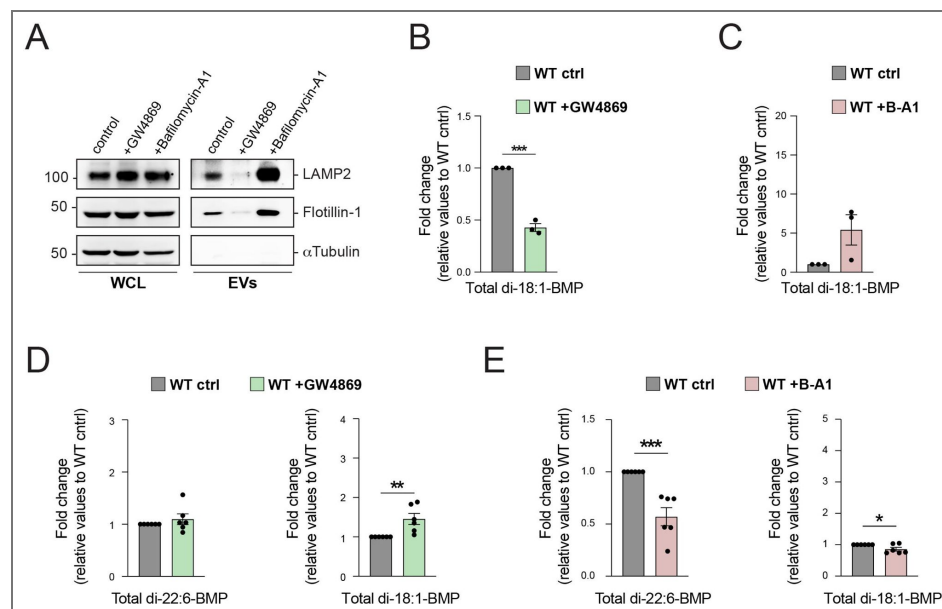


Figure 6. Pharmacological modulation of EV-mediated BMP exocytosis.

(A) Whole cell lysates (WCL) and isolated EVs from WT MEF cells treated with 10μM GW4869 or 10nM bafilomycin-A1 (B-A1) for 24h were analyzed by immunoblotting. Representative immunoblots of LAMP2, Flotillin-1 and α-Tubulin are shown. Molecular weight marker mobility is shown in kDa. (B-C) UPLC-MS/MS determination of BMP isoforms normalized to protein content from WT MEF cells treated with 10nM B-A1 (B) or 10μM GW4869 (C) for 24h. Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 6 independent experiments (mean ± SEM). Significance determined by two-tailed unpaired *t* test **p*<0.05, ***p*<0.01, ****p*<0.001, *****p*<0.0001. (D-E) Quantitation of BMP isoforms normalized to protein content in EVs isolated from MEF WT cells treated with 10nM B-A1 (D) or 10μM GW4869 (E) for 24h. Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 3 independent experiments (mean ± SEM). Significance determined by two-tailed unpaired *t* test **p*<0.05, ***p*<0.01, ****p*<0.001.

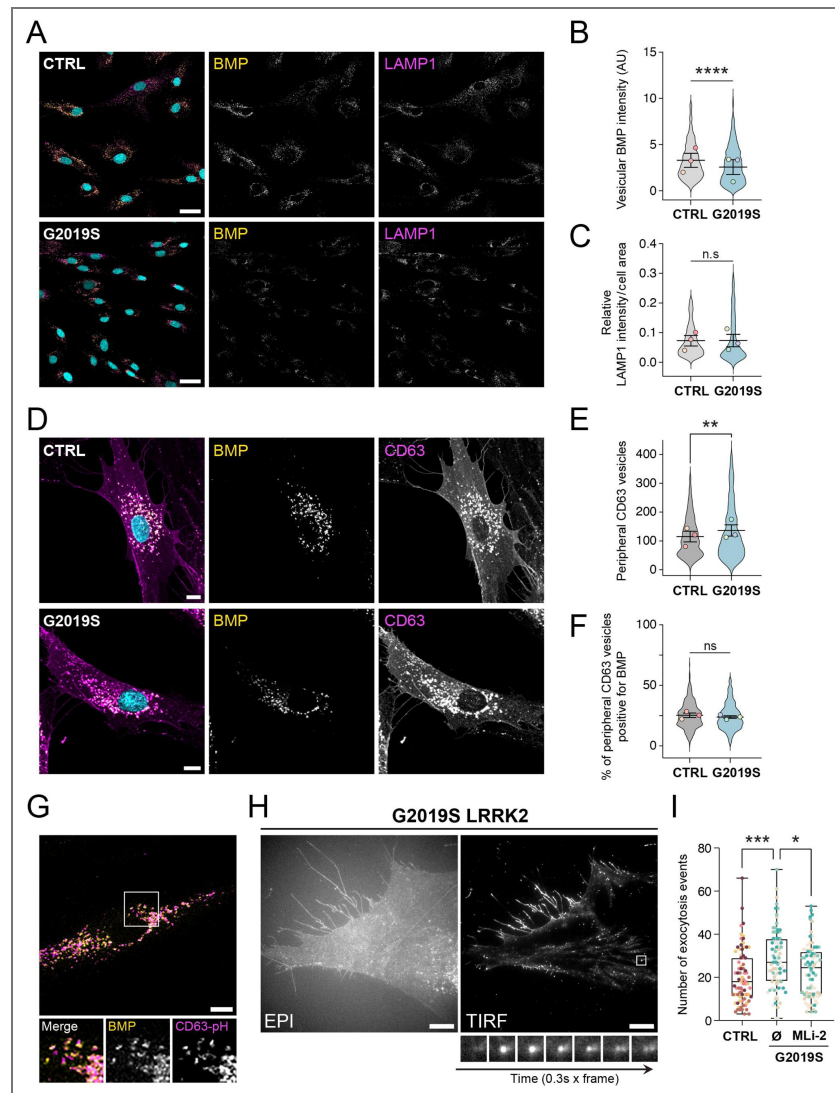


Figure 7. Patient-derived G2019S LRRK2 fibroblasts exhibit alterations of antibody-accessible BMP and increased endolysosomal exocytosis.

(A) Confocal microscopy of endogenous BMP (yellow) and LAMP1 (magenta) immunofluorescence in control (CTRL) and LRRK2-G2019S mutant-derived fibroblasts. Scale bar: 40 μ m. (B–C) Quantification of vesicular BMP intensity (B) and LAMP1 relative intensity (C) per cell area. Colored dots represent the mean value of three independent experiments ($n=3$ CTRL and $n=3$ G2019S LRRK2 fibroblast cell lines); violin plots show the distribution of individual cell data (60 cells per independent experiment). Significance determined by two-tailed paired t test; **** $p < 0.0001$. (D) Confocal microscopy of endogenous BMP (yellow) and CD63 (magenta) immunofluorescence in CTRL and G2019S LRRK2 fibroblasts. Scale bar: 10 μ m. (E–F) Quantification of the number of CD63 vesicles in the peripheral cell region (E) and the percentage of peripheral CD63 vesicles positive for BMP (F). Colored dots represent mean values of three independent experiments ($n=3$ CTRL and $n=3$ G2019S LRRK2 fibroblast cell lines); violin plots show the distribution of individual cell data (60 cells per independent experiment). Significance determined by two-tailed paired t test; ** $p < 0.01$. (G) A representative confocal microscopy image of endogenous BMP (yellow) and transduced CD63-pHluorin (magenta) in a CTRL human fibroblast cell line used for total internal reflection fluorescence (TIRF) microscopy experiments. Inset image shows co-localization between BMP and CD63-pHluorin in vesicular structures (a similar degree of co-localization was observed in G2019S LRRK2 cells; data not shown). Scale bar: 10 μ m. (H) Representative epifluorescence (EPI) and TIRF microscopy images of the same cell from a G2019S LRRK2 patient-derived fibroblast cell line treated with vehicle (DMSO) overnight. Inset images show a TIRF microscopy time-lapse sequence (0.3 s per frame) from a single pHluorin-CD63-positive fusion event at the plasma membrane. Scale bar: 10 μ m. (I) Quantification of pHluorin-CD63-positive fusion events in stably expressing control (CTRL) and G2019S LRRK2 fibroblasts treated with vehicle ($\emptyset$) or 200 nM MLI-2 for 16 h. Each dot represents one cell ($n=20$ cells quantified from four CTRL and four G2019S LRRK2 cell lines). Significance determined by Tukey's multiple comparisons test, ordinary one-way ANOVA; * $p < 0.05$, ** $p < 0.001$.

LRRK2 mutations exhibit alterations in antibody-accessible BMP comparable to those observed in MEFs, and that a pool of LRRK2 may be present in CD63/CLN5-positive endolysosomes likely enriched in BMP.

CD63-positive compartments have been implicated in exocytic processes rather than catabolic activities^{49–51}. In line with these observations, our data indicating that a pool of LRRK2 co-fractionates with isolated CD63-positive endolysosomes prompted us to further examine the intracellular distribution of CD63 in these cells. An increased number of CD63-positive vesicles near the plasma membrane was quantified in mutant LRRK2 fibroblasts compared to control cells (Figure 7D and E [↗](#)). On average, ~25% of CD63-positive peripheral endolysosomes were also positive for BMP, with no observable differences between control and mutant LRRK2 cells (Figure 7D and F [↗](#)). The peripheral enrichment observed in G2019S LRRK2 fibroblasts suggests enhanced recruitment of CD63-positive compartments toward exocytic sites at the cell surface, supporting the notion that these vesicles may eventually fuse with the plasma membrane to release EVs, as previously proposed^{49,50}.

To determine more precisely whether LRRK2 kinase activity modulates the exocytosis of CD63-positive endolysosomes, we performed live-cell total internal reflection fluorescence (TIRF) microscopy using CD63-pHluorin, a genetically encoded fluorescent reporter of exosome release^{47,50}. For these experiments, four independent control and G2019S LRRK2 patient-derived fibroblast cell lines were transduced to stably express CD63-pHluorin. The presence of BMP in CD63-pHluorin-positive endolysosomes was confirmed by immunofluorescence analysis after selection of transduced cells (Figure 7G [↗](#)). Consistent with recent findings in human induced pluripotent stem cell (iPSC)-derived neurons harboring the R1441H LRRK2 pathogenic mutation⁵², TIRF microscopy revealed an overall increase in endolysosomal exocytosis in the four G2019S LRRK2 fibroblast cell lines tested, which was significantly reduced upon MLI-2 treatment (Figure 7H and I [↗](#); Videos 1 and 2). Together, these findings are consistent with our previous observations in MEFs (Figures 2E [↗](#) and 3E [↗](#)) and support our initial hypothesis that LRRK2 kinase activity drives exocytosis and release of BMP-enriched EVs.

Discussion

Here we have shown that: (a) hyperactive LRRK2 kinase upregulates mass spectrometry-determined BMP levels and the extracellular vesicles (EVs) that release BMP in MEF cells. (b) Pharmacological inhibition of LRRK2 and GCase using MLI-2 and CBE, respectively, modulates BMP abundance in cells and isolated exosomal fractions in opposite directions without interfering with the kinetics of BMP biosynthetic or catabolic rates, although the effect of GCase inhibition on EV-associated BMP showed more variability across experiments. While PD-associated mutations in *LRRK2* increase kinase activity, *GBA1* pathogenic variants are associated with decreased GCase enzymatic function. Therefore, MLI-2 treatment represents a potential therapeutic rescue, whereas CBE phenocopies pathogenic conditions. (c) Although the hyperactive kinase mutation is associated with an overall increase in the BMP synthesis, neither MLI-2 or CBE affected BMP biosynthetic rate in the MEFs. (d) The expression of CLN5, a BMP synthase^{45,46}, is upregulated in mouse and patient-derived fibroblasts with PD-associated LRRK2 mutations, consistent with the lipidomics data indicating elevated amounts of cellular BMP. (e) BMP release was modulated by pharmacological agents known to modulate EV secretion. (f) Finally, G2019S LRRK2 human fibroblasts exhibit enhanced endolysosomal exocytosis and EV release which decreased upon MLI-2 treatment. Together, these data indicate that the previously reported increase in urinary BMP levels in LRRK2 mutation carriers reflects dysregulated exocytosis of BMP-containing EVs.

Pioneering studies by Gruenberg and colleagues underscore the importance of BMP in endolysosome homeostatic regulation and generation of endolysosomal ILVs^{14,18,19} that become exosomes upon release. Subsequent studies revealed that cellular levels of this atypical phospholipid are invariably altered in many neurodegenerative disorders characterized by endolysosome dysfunction²⁴. Recently, key lysosomal enzymes involved in distinct steps of the BMP biosynthetic pathway were identified, including CLN5 which plays a critical role in this process^{45,46}. Our microscopy analysis revealed decreased antibody-accessible BMP levels in

R1441G LRRK2 cells. In contrast, our targeted lipid pathway analysis measurements consistently showed higher total BMP levels in R1441G LRRK2 cells compared with WT cells. As we have reported previously⁴⁰, BMP antibody detects only a sub-pool of “accessible” BMP while mass spectrometry detects all pools. Alternately, not all BMP isoforms may be detected equally well. Given the essential roles of BMP in endolysosomal catabolism, our immunofluorescence data predict that LRRK2 mutant cells have defective degradative capacity, consistent with recent reports^{31,53}. In addition, in both MEFs and patient-derived mutant LRRK2 fibroblasts we found upregulation of the BMP-synthesizing enzyme CLN5 and, in MEFs, a concomitant increase in LAMP2 protein levels. These findings suggest that LRRK2 may participate in endolysosome biogenesis regulation. Indeed, in macrophages and microglia, LRRK2 regulates the levels of multiple lysosomal proteins by inhibiting TFEB and MiTF⁵³, transcription factors that activate lysosome-related gene expression by binding to coordinated lysosomal expression and regulation (CLEAR) elements⁵⁴. Although the promoter region of CLN5 contains a potential TFEB binding site⁵⁴, formal evidence for TFEB-mediated transcription of CLN5 is still lacking. The increase in CLN5 (and LAMP2 in MEFs) protein expression may also reflect lysosomal stress⁵⁴. Interestingly, MLI2 restored CLN5 levels in both murine-derived and human fibroblasts, further providing support that BMP is a pharmacodynamic biomarker of not just target modulation but also a potential disease-relevant lysosomal pathway modulation. Our findings of CLN5 upregulation in mutant LRRK2 MEFs, validated by similar results in patient-derived fibroblasts, further establish MEFs as a relevant cellular model system for studying LRRK2 kinase activity and its therapeutic targeting^{55–58}. Taken together, our data indicate that LRRK2 activity modulates specific components of the lysosomal network, including CLN5 and LAMP2. Nevertheless, quantitative analysis of additional endolysosomal markers (LAMP1 and CD63) in human fibroblasts did not reveal statistically significant differences between control and mutant LRRK2 cells. The elevated LAMP2 expression observed in the engineered MEF clone expressing R1441G may reflect a cell type-specific effect, potentially linked to differential penetrance of LRRK2 signaling on the lysosome biogenesis response.

We and others reported that BMP is aberrantly high in urine derived from LRRK2 and GBA mutation-carriers^{27–29}. These results have also been reproduced in animal models, in which urinary BMP levels decrease upon administration of MLI-2 and other LRRK2 kinase inhibitors^{41–43}. The present study complements these previous observations by providing evidence that EV-mediated BMP release can be regulated by LRRK2 and, although less consistently, by GCase enzymatic activity. Consistent with this model, recent studies have shown that aberrant LRRK2 and GCase activities influence EV secretion in cellular and animal models of PD, including human patients^{59–61}. Moreover, our data showing reduced exocytosis and EV release in mutant LRRK2 MEFs and patient-derived fibroblasts treated with MLI-2 is consistent with previous observations that LRRK2 kinase inhibition leads to surfactant accumulation in type 2 pneumocytes in the lung, likely as a consequence of impaired lysosome-related organelle exocytosis⁴². Moreover, a recent study also reporting decreased exocytosis of CD63-pHluorin-positive compartments in MLI-2-treated primary murine neurons and human induced pluripotent stem cell (iPSC)-derived neurons harboring pathogenic LRRK2 mutations⁵², further supports our conclusions.

BMP is enriched in urinary EVs⁶². However, other studies did not detect BMP enrichment in isolated EV fractions and concluded that BMP may be present in ILVs of endolysosomal subpopulations devoted to degradative activities rather than exocytic events^{63,64}, as previously proposed⁶⁵. Using our UPLC-MS/MS methodology, extracellular BMP can be efficiently detected in the urine of healthy and mutant LRRK2/GCase carriers^{27–29}, and additionally, in EV fractions derived from both WT and R1441G LRRK2 MEFs (this study). How does LRRK2 influence release of BMP-positive EVs (BMP-EVs)? A subset of Rab proteins that are master regulators of intracellular membrane trafficking pathways are important LRRK2 substrates⁵. Some of these Rabs include Rab10, Rab12 and Rab35, and have been shown to play regulatory roles in endolysosomal exocytosis and EV release^{66,67}. We thus speculate that phosphorylation of one or several of these Rabs may lead to enhanced BMP release. Indeed, G2019S LRRK2-mediated Rab35 phosphorylation has been proposed to promote EV-mediated α -synuclein release and propagation between cells^{68,69}. It is likely that aberrant LRRK2 and GCase activities trigger exocytosis, and associated

release of BMP-containing EVs, via a so-called clearance pathway activated due to accumulation of cytotoxic endolysosomal substrates^{70,71}. The contribution of these phosphorylated Rabs to EV-mediated BMP release will be of interest for future work.

In summary, this work provides evidence that pathological LRRK2 activity, and to a more variable degree GCase dysfunction, enhance EV-mediated BMP release without altering its metabolism in cells. The observed changes induced by MLI-2 highlight the potential of LRRK2 inhibition as a therapeutic strategy to restore endolysosomal homeostasis by reducing BMP production and aberrant EV release. What is the clinical significance of elevated urinary BMP levels in PD patients? Inhibition of LRRK2 improves endolysosomal function⁴³, consistent with increased LRRK2 and decreased GCase activities associated with pathogenic variants which worsen endolysosomal homeostasis^{6,13}; this may eventually lead to endolysosomal exocytosis followed by EV-mediated BMP release. We speculate that BMP-EVs may harbor a distinct molecular repertoire that could perhaps inform more precisely disease progression or pathobiology. Future investigations will be needed to study the contribution of dysfunctional LRRK2 and GCase in exocytosis of specific BMP-positive endolysosomal subpopulations and to further characterize the role of BMP-EVs in the context of disease pathophysiology but also as a PD diagnostic tool.

Materials and methods

Cell culture, antibodies and other reagents

Human-derived skin fibroblasts⁷⁴, wild type (WT) and R1441G LRRK2 mutant MEF cells were grown in Dulbecco's modified Eagle's media (DMEM) containing 10% fetal bovine serum, 2 mM L-glutamine, and penicillin (100 U/ml)/streptomycin (100 mg/ml). Cell lines were cultured at 37°C with 5% CO₂. Extracellular vesicle-free media was prepared by overnight ultracentrifugation of DMEM or RPMI supplemented with 10% FBS at 100,000 X g in a SW32Ti rotor. For bafilomycin-A1 (Sigma-Aldrich; catalog no. B1793) and GW4869 (SelleckChem; catalog no. S7609), the drug was added to cell media at 10 nM and 10 µM respectively and cells were cultured for 24 h before exosome collection. For MLI-2 (Tocris Bioscience; catalog no. 5756) and Condurotol β-Epoxyde (MERK; catalog no. 234599) were added to cell media at 200nM and 300µM, respectively. In our experiments, 48-hour incubations were necessary to sustain full LRRK2/GCase inhibition throughout the EV collection period. EV biogenesis, BMP biosynthesis, and packaging into EVs are time-dependent processes; therefore, extended incubation and collection periods (≥48 h) were required to allow downstream effects of LRRK2/GCase inhibition on BMP production and release to manifest, and to obtain sufficient EV material for biochemical and lipidomic analyses. 0.167mM Oleic acid (Cambridge Isotope Laboratories; catalog no. DLM-10012-0.001) and Docosahexaenoic acid (Cambridge Isotope Laboratories; catalog no. DLM-10012-0.001) were conjugated with 0.0278mM fatty acid-free BSA (Sigma-Aldrich; A8806) at a 1:6 molar ratio. Primary antibodies diluted in PBS with 1% BSA (for immunofluorescence) or 5% skim milk (for immunoblotting) were mouse anti-lysobisphosphatidic acid (anti-BMP) clone 6C4 1:1000 (EMD Millipore; catalog no. MABT837), rat monoclonal anti-mouse LAMP2 1:1000 (Developmental Studies Hybridoma Bank, catalog no. GL2A7), mouse anti-Flotillin-1 1:1000 (BD Biosciences; catalog no. 610821), rabbit anti-phospho-Rab10 1:1000 (Abcam; catalog no. ab230261), mouse anti-alpha tubulin 1:10000 (SigmaAldrich; catalog no. T5168), rabbit anti-CLN5 1:000 (Abcam; catalog no. ab170899), rabbit anti-phosphoLRRK2 1:000 (Abcam; catalog no. ab133450), rabbit anti-LAMP1 XP D2D11 1:1000 (Cell Signalling; catalog no. 9091), rabbit anti-CD63 [EPR22458-280] 1:500 (Abcam; catalog no. AB252919), rabbit anti-LRRK2 1:500 (Abcam; catalog. No. AB133474), mouse anti-calnexin (BD Transduction; 610523) 1:1000 and rabbit anti-GFP (Abcam, catalog no. AB290) 1:500.

Immunofluorescence

Wild type (WT) and R1441G LRRK2 mutant MEF cells were fixed with 3.7% (v/v) paraformaldehyde for 15 min, washed and permeabilized/blocked with 0.1% saponin/1% BSA-PBS before staining with anti-LAMP2 and anti-BMP antibodies followed by goat anti-rat AlexaFluor555 and donkey anti-mouse AlexaFluor488. Nuclei were stained using 0.1 mg/ml DAPI (SigmaAldrich; catalog no.

D9542). Coverslips were mounted on glass slides with Mowiol. Microscopy images were acquired using a Zeiss LSM880 laser scanning spectral confocal microscope (Carl Zeiss) equipped with an Axio Observer 7 inverted microscope, a blue diode (405 nm), argon (488 nm), diode-pumped solid-state (561 nm), and HeNe (633 nm) lasers, and a Plan Apochromat 63×/NA1.4 oil-immersion objective lens. DAPI, Alexa Fluor 488, Alexa Fluor 555 were acquired sequentially using 405-, 488- and 561-nm laser lines; Acousto-Optical Beam Splitter as a beam splitter; and emission detection ranges 415-480 nm, 500-550 nm, 571-625 nm, respectively. The confocal pinhole was set at 1 Airy unit. All images were acquired in a 1,024 × 1,024-pixel format. BMP and LAMP2 intensities were quantified using CellProfiler⁷⁵. All images for BMP-LAMP1 and BMP-CD63 analyses in human fibroblasts were acquired in tile-scan mode (5 × 5 field of view) with a z-step of 5, yielding a 2354 × 2354-pixel format. Tile-scan images were stitched with a 10% overlap, and maximum-intensity projections were generated using the Zeiss Zen Blue software. BMP and LAMP1 fluorescence intensities, as well as the number of peripheral CD63-positive vesicles and the percentage of peripheral CD63 vesicles positive for BMP, were quantified using Fiji (ImageJ) software.

Lentivirus production and cell transduction

To produce CD63-pHluorin lentivirus, a lentivirus vector containing the CD63-pHluorin-expressing cassette⁴⁷ was combined with third-generation lentiviral packaging mix (1:1:1 mix of pMD2.G, pRSV-Rev, and pMDLg/pRRE) at a 1:1 ratio and transfected into HEK293T cells using Polyethylenimine (PEI, #23966-2, PolySciences Inc.), as previously described⁷⁶. Transfected cells were grown for 72 hr and then the supernatant containing the viral particles was collected and passed through a sterile 0.45 µm Millex syringe filter (Millipore). Human-derived skin fibroblasts grown in 60 mm culture dishes were infected by adding a 1:1 mixture of complete DMEM medium and viral supernatant supplemented with 8 mg/mL polybrene, followed by a 24h incubation. The supernatant was removed, and cells were grown in fresh DMEM medium for 48-72 hr. Infected cells were selected by growing in DMEM medium supplemented with 1-2 mg/mL puromycin (Sigma- Aldrich) for 3-5 days. Cells stably expressing CD63-pHluorin were recovered in complete DMEM medium lacking puromycin for 48-72 hours.

Total Internal Reflection Fluorescence (TIRF) Live Microscopy

Control and G2019S LRRK2 human fibroblasts stably expressing pHluorin-CD63 were cultured on 25 mm coverslips and treated with DMSO (vehicle) or 200 nM MLI-2. After 16 h, cells were imaged using a Zeiss LSM880 Airyscan Elyra PS.1 laser scanning spectral confocal microscope (Carl Zeiss) equipped with an Axio Observer Andor1 inverted microscope. Imaging was performed with an α Plan-Apochromat TIRF 100×/1.46 NA oil DIC objective, a BP495–575 + LP750 filter set, and an sCMOS camera operated in TIRF mode. Excitation was achieved with an HR diode laser (488 nm, 200 mW). Time-lapse videos were acquired at 0.30 s per frame for a total of 540 frames per cell analyzed. All imaging experiments were performed at 37°C in a humidified incubator (5% CO₂). Fusion events for each cell were detected and quantified as sudden increases in fluorescence at the cell surface using the AMvBE macro in Fiji (ImageJ), as described⁷⁷. Twenty cells were imaged and quantified for each of the four fibroblast cell lines within each cohort.

Immunoblotting

Cells were lysed in lysis buffer (50 mM Hepes, 150 mM KCl, 1% Triton X-100, 5 mM MgCl₂, pH 7.4) supplemented with a protease/ phosphatase inhibitor cocktail (1mMNa₃VO₄, 10mMNaF, 1 mM PMSF, 10 µg/ml leupeptin, and 10 µg/ml aprotinin). After centrifugation at 12,000 g for 10 min, protein concentrations were measured in the cleared lysates. Equal protein amounts were loaded in lanes corresponding to whole-cell lysate (WCL) fractions and normalized based on α- Tubulin levels. For the extracellular vesicle (EV) fractions, all protein recovered from EV pellets after isolation was loaded. Exosome final pellets were resuspended in SDS-PAGE loading dye (50 mM Tris- HCl pH 6.8, 6% glycerol, 2% SDS, 0.2% bromophenol blue), heated briefly at 95°C, resolved by 10% SDS-PAGE gel and transferred onto nitrocellulose membranes (Bio-Rad Laboratories; catalog no. 1620115) using a Bio-Rad Trans-Blot system. Membranes were blocked with 5% skim milk in

TBS with Tween-20 for 60 min at RT. Primary antibodies were diluted in blocking buffer and incubated overnight at 4°C. HRP-conjugated secondary antibodies (Bio-Rad Laboratories) diluted in blocking buffer at 1:3000 were incubated for 60 min at RT and developed using EZ-ECL (Biological Industries) or SuperSignal West Femto (Thermo Scientific; catalog no. 34094). Blots were imaged using an ImageQuant LAS 4000 system (GE Healthcare) and quantified using ImageJ software.

Transmission Electron Microscopy

Cells in culture were washed in PBS and fixed for 1 h in 2.5% glutaraldehyde in 0.1 M phosphate buffer (PB) at RT. Next, samples were slowly and gently scraped and pelleted in 1.5 ml tubes. Pellets were washed in PB and incubated with 1% OsO₄ for 90 min at 4 °C. Then samples were dehydrated, embedded in Spurr and sectioned using Leica ultramicrotome (Leica Microsystems). Ultrathin sections (50-70 nm) were stained with 2% uranyl acetate for 10 min, a lead-staining solution for 5 min and observed using a transmission electron microscope, JEOL JEM- 1010 fitted with a Gatan Orius SC1000 (model 832) digital camera. Multivesicular Endosomes (MVE) and Intraluminal Vesicles (ILV) were identified by morphology and MVE area and ILV number were measured using ImageJ.

Extracellular vesicle isolation

MEF adherent cell cultures were seeded at $3\text{--}4 \times 10^6$ cells/plate in EV- free complete DMEM and grown for 24-48 h. Cell cultures were centrifuged at 300 g, 5 min to pellet cells, and supernatants were centrifuged again at 2100 g, 20 min to pellet dead cells. After this, filtration through a 0.22 µm filter unit (Millipore) was performed. Filtered media were ultracentrifuged for 75 min at 100,000 g in SW32Ti rotor to pellet extracellular vesicles. The supernatant was discarded, and the pellet (small EVs) re-suspended in phosphate-buffered saline (PBS) and centrifuged again for 75 min at 100,000 g in 140AT rotor. exosome pellets were frozen and stored at -80°C prior to further processing. In all EV-related experiments, we seeded the same number of EV-producing cells per condition, and the resulting EV-derived data (from both immunoblotting and lipidomics analyses) were normalized to the corresponding whole cell lysate (WCL) protein content to ensure comparability across conditions.

Lipidomic analysis

Targeted quantitative ultra-performance liquid chromatography tandem mass spectrometry was used to accurately quantitate the three geometrical isomers (2,2-, 2,3- and 3,3-) of di-22:6-BMP and di-18:1-BMP in control or treated MEF cells. Lipidomic analyses were conducted by Nextcea, Inc. as previously described⁶⁶. Standard curves were prepared using authentic BMP reference standards. Protein was determined by bicinchoninic acid protein assay. A multiplexed quantitative ultraperformance liquid chromatography tandem mass spectrometry method was used to simultaneously quantitate sample glucosylceramides (GluCer d18:1/16:0, d18:1/18:0, d18:1/22:0, d18:1/24:0, d18:1/24:1), galactosylceramides (GalCer d18:1/16:0, d18:1/18:0, d18:1/22:0, d18:1/24:0, d18:1/24:1), glucosylsphingosine (GluSph 18:1), and galactosylsphingosine (GalSph 18:1). Standard curves were prepared from related standards using a class-based approach. Internal standards were used for each analyte reported. A SCIEX Triple Quad 7500 mass spectrometer was used in positive electrospray ionization mode for detection (SCIEX, Framingham, MA, USA). Injections were made using a Shimadzu Nexera XR UPLC (ultraperformance liquid chromatography) system (Shimadzu Scientific Instruments, Kyoto, Japan). The instruments were controlled by SCIEX OS 2.0 software.

Metabolic labeling

WT and R1441G LRRK2 mutant MEF cells were treated with 200nM MLI-2 or 300µM CBE for 16h. Cells were washed twice with PBS and incubated in serum-free DMEM ± MLI-2 or CBE for 1h. Next, Incubate cells in 0.167mM ¹³C-labeled Oleic Acid (OA) and 0.167mM deuterated docosahexaenoic acid (DHA) in serum-free DMEM containing 0.0278mM BSA for 25 min. Cells were washed with

serum-free DMEM and chased with serum-free DMEM for short time points (0, 15, 30, 45, 60min) or complete DMEM for long timepoints (8, 24 and 48h). Cells were washed with PBS, trypsinized and collected in cold PBS. Cells were pelleted by centrifugation at 1000rpm for 7min and stored at -80°C prior to further processing.

GCase activity

GCase activity was measured by using the fluorogenic substrate PFB-FDGlu (Invitrogen, Carlsbad, CA). WT and R1441G LRRK2 mutant MEF cells were treated with 300μM CBE for 24h. After this, 150 μg/ml PFB-FDGlu (Thermo Fisher) was then added to cells followed by incubation for 2h at 37°C. Cells were then washed twice with PBS, trypsinized and fixed in cold 2% paraformaldehyde-PBS for 30 min. Cells were washed twice in PBS before assayed on a FACS analyzer (FACScan; BD) equipped with a 488 nm laser and 530 nm-FITC filter. Data were analyzed using Flowjo software (Tree Star).

Subcellular fractionation

Endolysosomes were isolated by ultracentrifugation of discontinuous sucrose density gradients as previously described⁷⁸. Briefly, control and G2019S LRRK2 fibroblasts were collected using a homogenization buffer containing 3mM Imidazole, 250nM sucrose, pH 7.4 and proteases inhibitors. Cell pellets were homogenized by 90 passages through a 27-gauge needle. Complete homogenization was assessed by phase-contrast microscopy. After this, samples were centrifuged for 10 min at 1000 g. The Post-nuclear supernatant (PNS) fractions were collected and adjusted to 40,2% (w/v) sucrose. PNS fractions were then loaded at the bottom of a micro- ultracentrifuge tube for Himac P70AT(RP70AT) Fixed Angle Rotor. Then, 35% sucrose, 25% sucrose, and 8% sucrose layers were poured stepwise over the PNS bottom layer. The discontinuous sucrose gradient was centrifuged for 3 h at 100,000 g using a Himac CS-(F)NX (Eppendorf) micro- ultracentrifuge. After this, 15 fractions were collected, and protein was trichloroacetic acid (TCA)- precipitated before SDS-PAGE analysis.

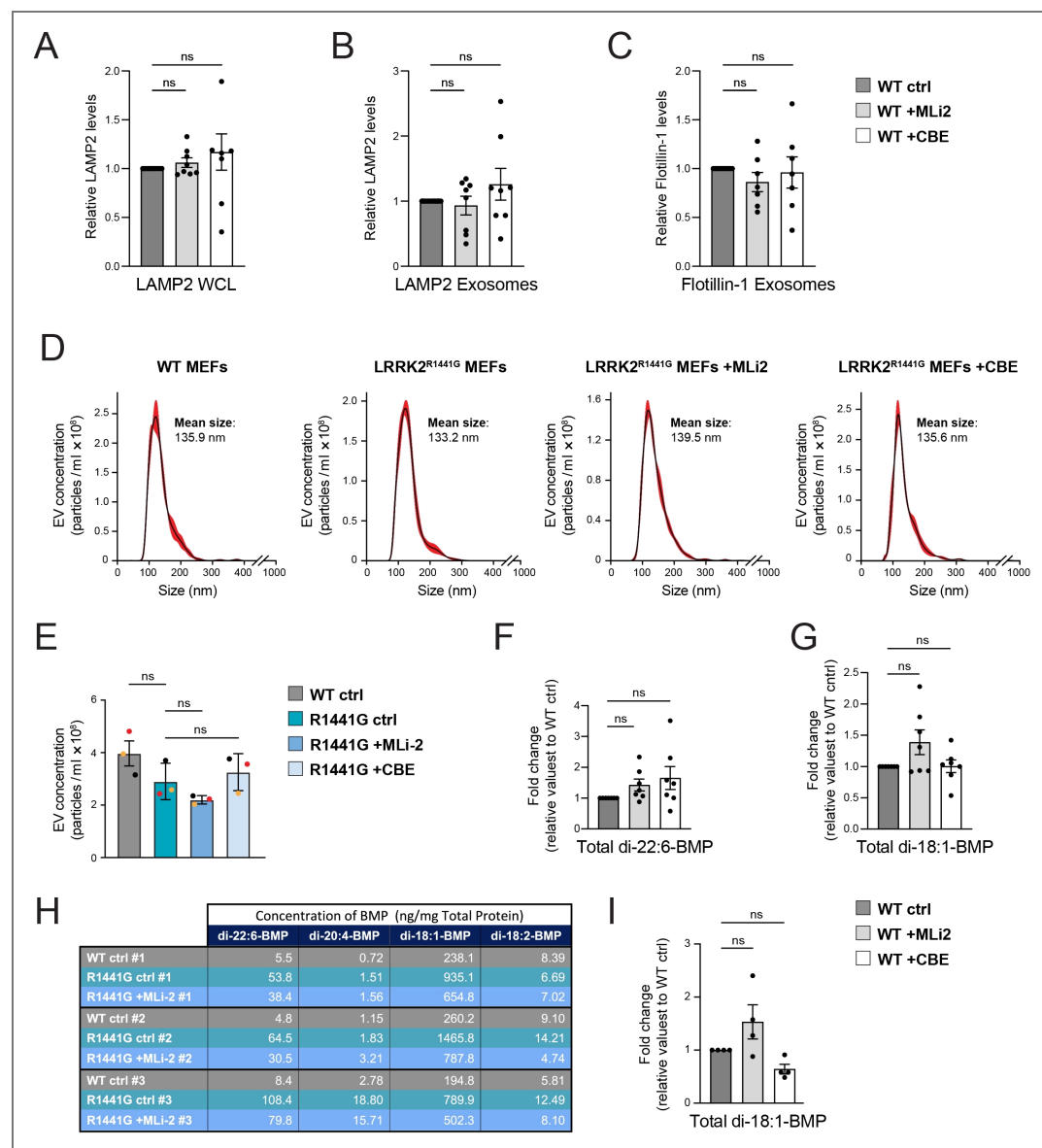
Statistics

Results are expressed as the mean ± SEM. Means were compared using Student's t test when two experimental conditions were independently compared. Unless otherwise noted, statistical significance between multiple comparisons was assessed via one-way or two-way ANOVA with uncorrected Fisher's LSD test using GraphPad Prism version 9.

Data availability

Data presented herein will be made available upon request. No large datasets were generated in this study.

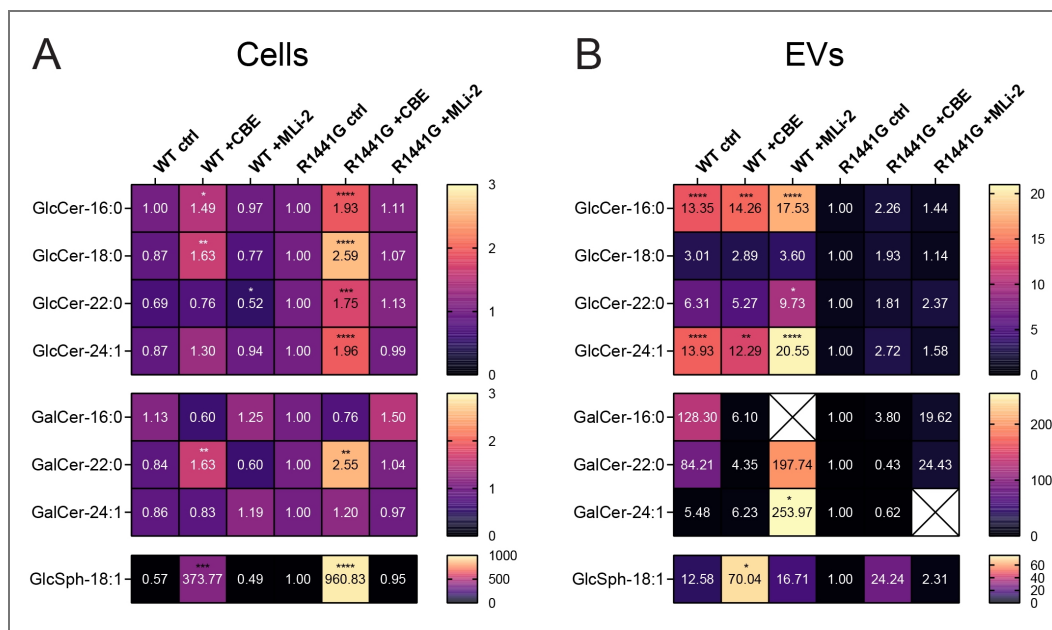
Supplemental figures



Supplemental Figure 1. Further characterization of MEF-derived EV fractions. (A-C) No significant differences in EV release between MLI-2/CBE-treated and untreated WT MEF cells. Quantification of LAMP2 and Flotillin-1 levels relative to WT control MEF cells in WCL (A) and isolated EVs (B-C). Data from 7-8 independent experiments (mean ± SEM). Significance determined by ordinary one-way ANOVA, uncorrected Fisher's LSD. (D) and (E), Characterization of WT or R1441G LRRK2 MEF-derived purified EVs by Nanoparticle Tracking Analysis (NTA). (D) Representative plots of EV size distribution in each indicated condition. (E) Yield comparison of MEF-derived EVs from each indicated condition determined by NTA. Each colored dot represents an independent experiment. (F,G) No significant differences in cellular BMP levels between MLI-2/CBE-treated and untreated WT MEFs, as measured by UPLC-MS/MS. (H) Table showing absolute values of UPLC-MS/MS measurements of cellular docosahexaenoyl (22:6)-, arachidonoyl (20:4)-, oleoyl (18:1)-, and linoleyl (18:2)-BMP species from 3 representative independent experiments. (I) No significant differences in EV-associated BMP levels between MLI-2/CBE-treated and untreated WT MEF cells, as measured by UPLC-MS/MS. Lipidomics measurements of cellular (F,G) and EV-associated (I) BMP isoforms normalized to protein content from cells treated with 200nM MLI-2 or 300μM CBE for 48h, or left untreated (ctrl). Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 6-7 independent experiments (mean ± SEM). Significance determined by ordinary two-tailed paired *t* test; ns, not significant.

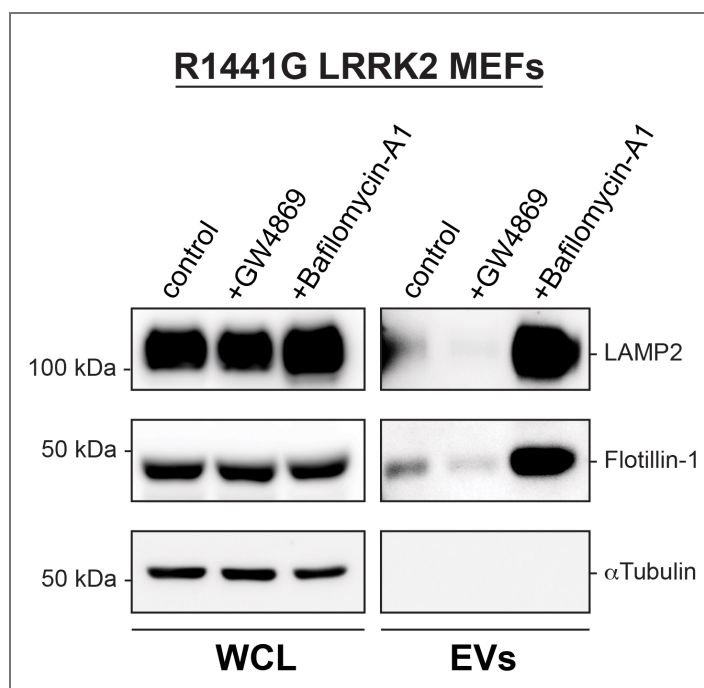
Supplemental Figure 2. Quantitative analysis of GCase substrates in MEF cells and EV fractions.

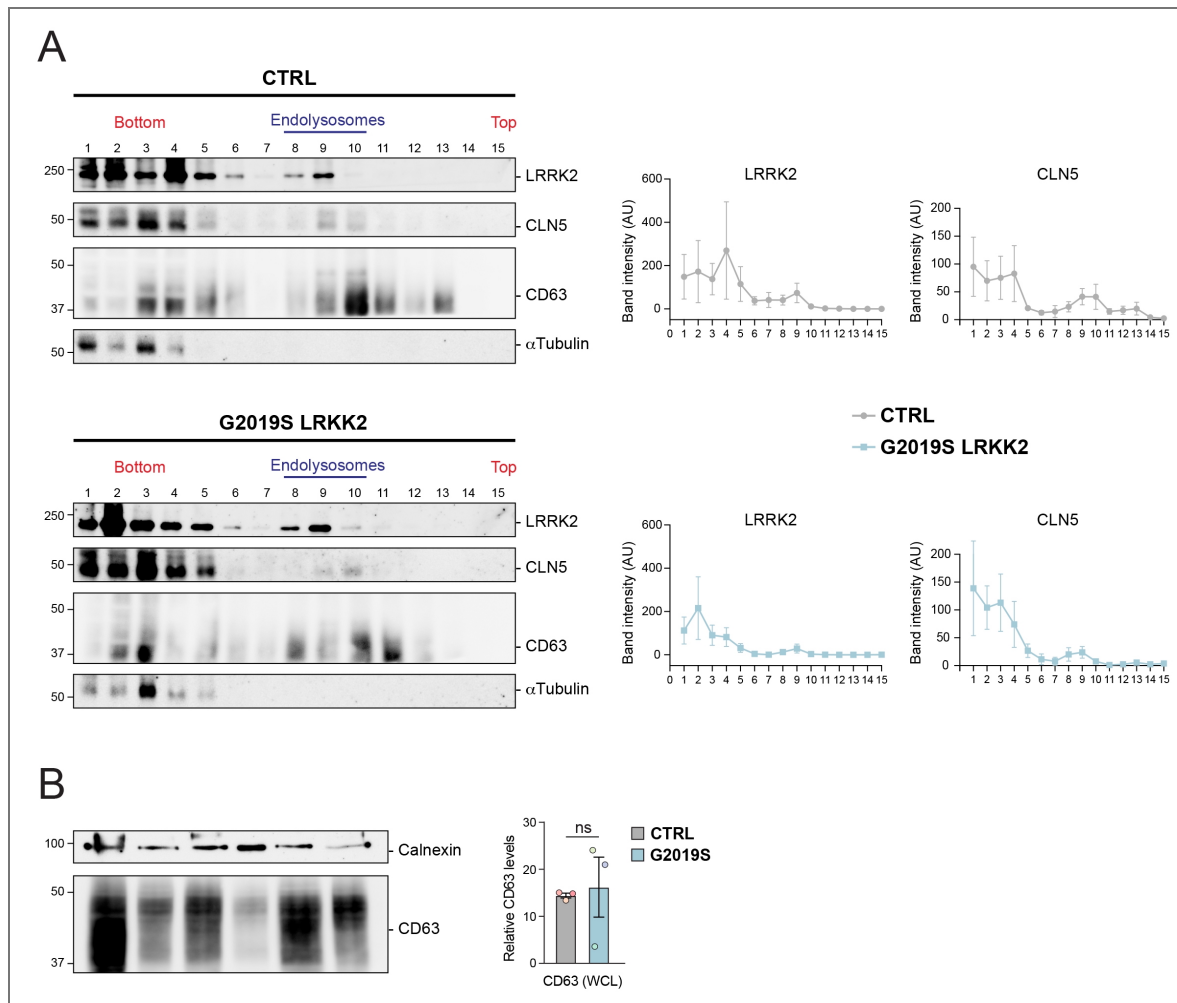
(A-B) Lipidomic determination of Glucosylceramide (GlcCer), Galactosylceramide (GalCer) and Glucosylsphingosine (GlcSph) isoforms normalized to protein content from WT and R1441G LRRK2 mutant MEF cells **(A)** and isolated EVs **(B)** treated with 300μM CBE or 200nM MLI-2 for 48h. Heatmap showing values as fold change relative to R1441G LRRK2 mutant MEF cell control. Only GlcCer, GalCer and GlcSph isoforms that were detected are shown. Data from 3-9 independent experiments (mean ± SEM). Significance determined by ordinary two-way ANOVA, uncorrected Fisher's LSD *p<0.05, **p<0.01, ***p<0.001 ****p<0.0001.



Supplemental Figure 3. Pharmacological modulation of EV-mediated BMP exocytosis in mutant LRRK2 MEF cells.

Whole cell lysates (WCL) and isolated EVs from R1441G LRRK2 MEF cells treated with 10μM GW4869 or 10nM bafilomycin-A1 for 24h were analyzed by immunoblotting. Representative images of LAMP2, Flotillin-1 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa.





Supplemental Figure 4. Analysis of endolysosomal fractions in control and G2019S LRRK2 human- derived skin fibroblasts.

(A) Immunoblots showing fractions recovered after ultracentrifugation of sucrose gradients of cell extracts from control (CTRL) and G2019S LRRK2 mutant-derived fibroblasts. Endolysosomes fractions correspond to fractions 8-10. Representative immunoblots for LRRK2, CLN5, CD63 and α -tubulin are shown. Plots on the right show quantification of LRRK2 and CLN5 band intensities. $n=3$ CTRL and $n=3$ G2019S LRRK2 fibroblast cell lines. Molecular weight marker mobility is shown in kDa. (B) Whole cell lysates from CTRL individuals and G2019S LRRK2 mutant-derived fibroblasts analyzed by immunoblotting. Representative blots of CD63 and calnexin are shown. Plot on the right shows quantification of relative CD63 protein levels in whole-cell lysates normalized to Calnexin (used as a loading control). Molecular weight marker mobility is shown in kDa. Significance determined by ordinary two-tailed paired t test; ns, not significant.

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Additional information

Author contributions

K.M.M. and A.L. conceived the project with input from F.H. All experiments were carried out by E.M-S. and A.L. except for mass spectrometric analysis of BMP (F.H), and CLN5 biochemical analysis in human fibroblasts (M.C.). R.F.S., M.E., A.G and M.J.M provided patient-derived fibroblasts. C.E. captured and analyzed transmission electron microscopy images. K.M., A.L., S.R.P. and F.H. oversaw the project. K.M. and A.L. obtained research funding. A.L. wrote and edited the manuscript with critical revision from all co-authors.

Abbreviations

BMP, bis(monoacylglycerol)phosphate; CBE, conduritol β -epoxide; DHA, docosahexaenoic acid; GCase, glucocerebrosidase; ILV, intralumenal vesicle; LBPA, Lysobisphosphatidic acid; LRRK2, leucine-rich repeat kinase 2; PD, Parkinson's disease; WT, wild type

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Additional files

Video 1. [↗](#) Time-lapse of vehicle (DMSO)-treated human G2019S LRRK2 fibroblast cell stably expressing CD63-pHluorin. Timestamp on upper left corner indicates seconds. Fusion events are indicated by white circles. Scale bar: 10 μ m. Images were acquired at ~3.33 frames per sec.

Video 2. [↗](#) Time-lapse of MLI-2-treated human G2019S LRRK2 fibroblast cell stably expressing CD63-pHluorin. Timestamp on upper left corner indicates seconds. Fusion events are indicated by white circles. Scale bar: 10 μ m. Images were acquired at ~3.33 frames per sec.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Even though mutations in LRRK2 and GBA1 (which encodes the protein GCase) increase the risk of developing Parkinson's disease (PD), the specific mechanisms driving neurodegeneration remain unclear. Given their known roles in lysosomal function, the authors investigate how LRRK2 and GCase activity influence the exocytosis of the lysosomal

lipid BMP via extracellular vesicles (EVs). They use fibroblasts carrying the PD-associated LRRK2-R1441G mutation and pharmacologically modulate LRRK2 and GCase activity.

Strengths:

The authors examine both proteins at endogenous levels, using MEFs instead of cancer cells. The study's scope is potentially interesting and could yield relevant insights into PD disease mechanisms.

Weaknesses:

Many of the authors' conclusions are overstated and not sufficiently supported by the data. Several statistical errors undermine their claims. Pharmacological treatment is very long, leading to potential off target effects. Additionally, the authors should be more rigorous when using EV markers.

Comments on revisions:

The authors have not addressed most of my concerns. For example, instead of trying with a 1-2 hour MLi2 treatment, they cited all the papers that use extremely long time points for LRRK2 inhibition; the fact that other groups do it does not mean it is biologically correct. They also refused to quantify their western blots in a proper manner, without the "hyper-normalization" claiming that it is an accepted way to quantify western blots. Again, it is statistically incorrect and biologically impossible. They also do not have a satisfactory explanation as to why the R1441G cells (which increase LRRK2 kinase activity) have no effect on EV release, but they still claim it is LRRK2 kinase activity dependent.

Overall, I am very confused by the model proposed by the authors. They only see increased EV release in the G2019S expressing cells, but not the R1441G cells, yet they claim that the increase of EV release is LRRK2 kinase activity dependent. Then, they claim that the presence of BMP (unchanged in R1441G vs CTL) in EVs is also LRRK2 kinase activity dependent. Finally, they perform TIRF with pHluorin-CD63 construct and observed an increase in G2019S cells vs CTL "further confirming that BMP release is associated with EV secretion". First, I could not see the increase in BMP release in G2019S cells (if I missed it, I apologize). And second, why didn't they do this experiment in R1441G cells? As, the R1441G cells have not displayed an increase in EV release compared to CTL cells, it could also be possible that the BMP release might be more abundant through lysosomal exocytosis (which could explain the pHluorin results) than EVs. Overall, the authors nicely demonstrate that the R1441G cells have more BMP species, likely due to increase CLN5 expression, but the release of the BMP is still not clear to this reviewer.

<https://doi.org/10.7554/eLife.106330.2.sa2>

Reviewer #2 (Public review):

Summary:

In this paper, authors used MEFs expressing the R1441G mutant of leucine-rich repeat kinase 2 (LRRK2), a mutant associated with the early onset of Parkinson's disease. They report that in these cells LAMP2 fluorescence is higher but BMP fluorescence is lower, MVE size is reduced and that MVEs contain less ILVs. They also report that LAMP2-positive EVs are increased in mutant cells in a process sensitive to LRRK2 kinase inhibition but are further increased by glucocerebrosidase (GCase) inhibition, and that total di-22:6-BMP and total di-18:1-BMP are increased in mutant LRRK2 MEFs compared to WT cells by mass spectrometry. They also report that LRRK2 kinase inhibition partially restores cellular BMP levels, and that GCase inhibition further increased BMP levels, and that in EVs from the LRRK2 mutant, LRRK2 inhibition decreases BMP while GCase inhibition has the opposite effect. Moreover, they

report that BMP increase is not due to increased BMP synthesis, although authors observe that CLN5 is increased in LRRK2 mutant cells. Finally, they report that GW4869 decreases EV release and exosomal BMP, while bafilomycin A1 increases EV release. They conclude that LRRK2 regulates BMP levels (in cells) and release (via EVs). They also conclude that the process is modulated by GCase in LRRK2 mutant cells, and that these studies may contribute to the use of BMP-positive EVs as a biomarker for Parkinson's disease and associated treatments.

Strengths:

This is a potentially interesting paper,. However, I had comments that authors needed to address to clarify some aspects of their study.

Weaknesses:

(1) The authors seem to have missed the point in their reply to my first comment. They mention the paper by Stuffers et al., who reports that endosome biogenesis continues without ESCRT. This is a nice paper, but it is irrelevant to the subject at hand. In my initial comment, I drew the author's attention to an apparent contradiction: higher LAMP2 staining in R1441G LRRK2 knock-in MEFs and yet smaller MVEs with a reduced surface area. LAMP2 being one of the major glycoproteins of MVE's limiting membrane, one would have expected lower LAMP2 staining if cells contain fewer and smaller MVEs. Authors now state that elevated LAMP2 expression in cells expressing R1441G reflects a cell type-specific effect (differential penetrance of LRRK2 signaling on lysosomal biogenesis), because amounts of LAMP1 and CD63 are similar in cells from LRRK2 G2019S PD patients and control cells (new Fig 7A-F). However, authors still conclude that LRRK2 modulates the lysosomal network, including LAMP2 and CLN5. Does it?

Similarly, the mass spec analysis of BMP (Fig S1H) does not support the data in Fig 1. Does this Table include all major isoforms found in these cells? If so, the dominant isoform is by far the di-18:1 isoform in wt and R1441G cells (at least 10X more abundant than other isoforms). Now, di-18:1-BMP is roughly 4X more abundant in R1441G cells when compared to wt cells, while BMP is reduced by half in R1441G cells (light microscopy in Fig 1). Authors argue that light microscopy may only detects a so-called antibody accessible pool. What is this? And why would this pool decrease in R1441G cells when LAMP2 is higher? Alternatively, they argue that the anti-BMP antibody may be less specific and detect other analytes. As I had already mentioned, this makes no sense, since the observed signal is lower and not higher. If authors do not trust their light microscopy analysis, why show the data?

(2) Cells contain 3 LAMP2 isoforms. Which one is upregulated and/or secreted in exosomes?

(3) The new Fig S4A is far from convincing. How were cells fractionated and what are the gradients (not described in Methods)? CD63 (presumably endolysosomes) is spread over fractions 8 - 13. LRRK2 (fractions 8-9) does not copurify with CD63. The bulk of LRRK2 is at the bottom (presumably cytosol if this is a floatation gradient), and a minor fraction moves into the gradient. CLN5 is even less clear since the bulk is also at the bottom with a tiny fraction only between LRRK2 and CD63. Also, why do authors conclude that a considerable pool of newly synthesized CLN5 did not reach its final destination at the endolysosome and may instead be retained in the ER? Where is the ER on the gradient?

(4) Fig S4B shows blots of whole cell lysates from CTRL and LRRK2 mutant-derived fibroblasts: 6 lanes are shown but without captions, containing varying amounts of calnexin and CD63. In addition, the blots look very dirty. Where is CD63? Is it the minor band at ~37 kD (as in Fig S4A)? Or the major band below the 50kD marker? What are the other bands on these blots? As a result, the quantification shown in the bar graph does not mean much.

(5) The cell content of 18:1-BMP is increased approx. 5X by BafA1 (Fig 6C) but amounts of 18:1-BMP secreted in EVs hardly changes (Fig 6E). Since BMP is mostly present as 18:1 isoform (22:6-BMP being only a minor species, Fig S1H), does it mean that BafA1 does not increase BMP secretion and/or only a minor fraction of total cellular BMP is secreted in exosomes?

Comments on revisions:

How come 0.2 mmol/L of 22:6 and 18:1 fatty acid both correspond to 65 µg/mL (Fig 4A)?

It is stated in the Legend of Fig4 that long (B-C) and short (D) chase time points are shown as fold change. There is no panel D in the figure.

<https://doi.org/10.7554/eLife.106330.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This useful study presents the potentially interesting concept that LRRK2 regulates cellular BMP levels and their release via extracellular vesicles, with GCase activity further modulating this process in mutant LRRK2-expressing cells. However, the evidence supporting the conclusions remains incomplete, and certain statistical analyses are inadequate. This work would be of interest to cell biologists working on Parkinson's disease.

Reviewer #1 (Public review):

Summary:

Even though mutations in LRRK2 and GBA1 (which encodes the protein GCase) increase the risk of developing Parkinson's disease (PD), the specific mechanisms driving neurodegeneration remain unclear. Given their known roles in lysosomal function, the authors investigate how LRRK2 and GCase activity influence the exocytosis of the lysosomal lipid BMP via extracellular vesicles (EVs). They use fibroblasts carrying the PD-associated LRRK2-R1441G mutation and pharmacologically modulate LRRK2 and GCase activity.

Strengths:

The authors examine both proteins at endogenous levels, using MEFs instead of cancer cells. The study's scope is potentially interesting and could yield relevant insights into PD disease mechanisms.

Weaknesses:

Many of the authors' conclusions are overstated and not sufficiently supported by the data. Several statistical errors undermine their claims. Pharmacological treatment is very long, leading to potential off-target effects. Additionally, the authors should be more rigorous when using EV markers.

We thank the reviewer for these valuable observations. In the revised manuscript, we have addressed each of these points as follows:

(1) Conclusions and data support – We carefully revised our text throughout the manuscript to ensure that all conclusions are better supported by the presented data. For instance, we now explicitly state that while pharmacological modulation supports the regulatory role of

LRRK2 activity in EV-mediated BMP release, we have softened our conclusions concerning the contribution of GCase in this model (see revised Results and Discussion sections).

(2) Statistical analyses – We reanalyzed experiments involving more than two groups and replaced simple t-tests with non-parametric Kruskal-Wallis tests followed by Dunn's post hoc comparisons. This approach, described in the updated figure legends (e.g., Figure 2D-F and H-J), provides a more rigorous statistical framework that accounts for small sample sizes and variability typical of EV quantifications.

(3) Pharmacological treatment duration – Prolonged MLi-2 treatments have been extensively used in the field without evidence of significant off-target effects. Several studies, including Fell et al. (2015, J Pharmacol Exp Ther 355:397-409), De Wit et al. (2019, Mol Neurobiol 56:5273-5286), Ho et al. (2022, NPJ Parkinson's Dis 8:115), Tengberg et al. (2024, Neurobiol Dis 202:106728), and Jaimon et al. (2025, Sci Signal 18:eads5761), have applied long-term (24-48 h) MLi-2 treatments at comparable concentrations without detecting toxicity or off-target alterations, including in MEFs (Ho et al., 2022; Dhekne et al., 2018, eLife 7:e40202). In our study, 48-hour incubations were necessary to sustain full LRRK2 inhibition throughout the extracellular vesicle (EV) collection period. EV biogenesis, BMP biosynthesis, and packaging into EVs are timedependent processes; therefore, extended incubation and collection periods (48 h) were required to allow downstream effects of LRRK2 inhibition on BMP production and release to manifest, and to obtain sufficient EV material for biochemical and lipidomic analyses. This experimental design also reflects our and others' previous observations in humans and non-human primates, where urinary BMP changes are associated with chronic or subchronic LRRK2 inhibitor treatment (Baptista MAS, Merchant K, et al. Sci Transl Med. 2020, 12:eaav0820; Jennings D, et al. Sci Transl Med. 2022, 14:eabj2658; Maloney MT, et al. Mol Neurodegener. 2025, 20:89). Importantly, under these conditions, we did not observe significant changes in cell viability or morphology, supporting that the treatment was well tolerated. We have clarified this rationale in the revised Methods section to emphasize that the prolonged incubation reflects the experimental design for EV isolation rather than a requirement for achieving LRRK2 inhibition.

(4) EV markers – We and others have reported enrichment of Flotillin-1 and LAMP proteins in isolated small EV fractions (Kowal et al., 2016; Lu et al., 2018; Mathieu et al., 2021; Ferreira et al., 2022). Moreover, LAMP proteins have been reported to be more enriched in EVs of endolysosomal origin (Mathieu et al., 2021). To further strengthen this point, we performed new experiments using a CD63-pHluorin sensor combined with TIRF microscopy, which allowed real-time visualization of CD63-positive exosome release. These new data (now presented in Figure 7, Panels G-I; Videos 1 and 2) confirm increased CD63-positive EV release in LRRK2 mutant fibroblasts, which was reversed by LRRK2 inhibition with MLi-2. The CD63-positive compartment was also largely BMPpositive (new Figure 7D, F, G), reinforcing our conclusions and providing additional rigor in EV marker validation.

Reviewer #2 (Public review):

Summary:

In this paper, the authors used MEFs expressing the R1441G mutant of leucine-rich repeat kinase 2 (LRRK2), a mutant associated with the early onset of Parkinson's disease. They report that in these cells LAMP2 fluorescence is higher but BMP fluorescence is lower, MVE size is reduced, and that MVEs contain less ILVs. They also report that LAMP2-positive EVs are increased in mutant cells in a process sensitive to LRRK2 kinase inhibition but are further increased by glucocerebrosidase (GCase) inhibition, and that total di-22:6-BMP and total di-18:1-BMP are increased in mutant LRRK2 MEFs compared to WT cells by mass spectrometry. They also report that LRRK2 kinase inhibition partially restores cellular BMP levels, and that GCase inhibition further increases BMP levels, and that in EVs from the LRRK2 mutant, LRRK2 inhibition decreases BMP while GCase

inhibition has the opposite effect. Moreover, they report that the BMP increase is not due to increased BMP synthesis, although the authors observe that CLN5 is increased in LRRK2 mutant cells. Finally, they report that GW4869 decreases EV release and exosomal BMP, while bafilomycin A1 increases EV release. They conclude that LRRK2 regulates BMP levels (in cells) and release (via EVs). They also conclude that the process is modulated by GCase in LRRK2 mutant cells, and that these studies may contribute to the use of BMP-positive EVs as a biomarker for Parkinson's disease and associated treatments.

Strengths:

This is an interesting paper, which provides novel insights into the biogenesis of exosomes with exciting biomedical potential. However, I have comments that authors need to address to clarify some aspects of their study.

Weaknesses:

(1) The intensity of LAMP2 staining is increased significantly in cells expressing the R1441G mutant of LRRK2 when compared to WT cells (Figure 1C). Yet mutant cells contain significantly smaller MVEs with fewer ILVs, and the MVE surface area is reduced (Figure 1D-F). This is quite surprising since LAMP2 is a major component of the limiting membrane of late endosomes. Are other proteins of endo-lysosomes (eg, LAMP1, CD63, RAB7) or markers (lysotracker) also decreased (see also below)?

As referenced in our original manuscript, several previous studies have reported endolysosomal morphological and homeostatic defects in cells harboring pathogenic LRRK2 mutations. LAMP2 can be upregulated as part of a lysosomal biogenesis or stress response (e.g., via MiT/TFE transcription factors such as TFEB; Sardiello et al., Science 2009, 325:473-477), whereas ILV biogenesis is primarily controlled by ESCRT- and SMPD3-dependent pathways that are regulated independently of MiT/TFE-driven transcriptional programs. Indeed, Stuffers et al. (Traffic 2009, 10:925-937) demonstrated that depletion of key ESCRT subunits markedly inhibited ILV formation while concomitantly increasing LAMP2 expression, highlighting the mechanistic dissociation between LAMP2 abundance and ILV number. In our study, we observed a similar pattern in R1441G LRRK2 MEFs, in which elevated LAMP2 staining and protein levels occurred despite a reduction in MVE size and ILV number. We interpret this as a compensatory lysosomal biogenesis response.

Our revised manuscript now includes new immunofluorescence data for BMP, LAMP1 and CD63 (New Figure 7, Panels A-F) together with biochemical analysis of CD63 protein levels (New Supplemental Figure 4, Panel B) in human skin fibroblasts derived from healthy donors and LRRK2 G2019S PD patients. Quantitative analysis of these experiments revealed no statistically significant differences in total cellular levels of either LAMP1 or CD63 between groups. However, we observed a consistent decrease in BMP immunostaining intensity (New Figure 7, Panel A and B), in agreement with our findings in mouse fibroblasts. We therefore propose that the elevated LAMP2 expression observed in the engineered MEF clone expressing R1441G may reflect a cell type-specific effect, potentially linked to differential penetrance of LRRK2 signaling on the lysosomal biogenesis response. We have updated the Results and Discussion section of the manuscript to incorporate and clarify these findings.

(2) LRRK2 has been reported to interact with endolysosomal membranes. Does the R1441G mutant bind LAMP2- and/or BMP-positive membranes?

We agree that LRRK2 has been reported to associate dynamically with endolysosomal membranes, particularly under conditions of endolysosomal stress or damage (Eguchi T, et al. PNAS 2018, 115:E9115-E9124; Bonet-Ponce L, et al. Sci Adv. 2020, 6:eabb2454; Wang X, et al. Elife. 2023, 12:e87255).

Nevertheless, to explore whether LRRK2 associates with BMP-positive endolysosomes, we performed subcellular fractionation followed by biochemical analysis of endolysosomal fractions, since our available LRRK2 antibodies did not provide reliable immunofluorescence signals. These experiments were carried out using human skin fibroblasts derived from both healthy controls and Parkinson's disease patients carrying the LRRK2-G2019S mutation. In both control and mutant fibroblasts, a pool of LRRK2 was detected in fractions positive for the BMP synthase CLN5 and the endolysosomal marker CD63 (New Supplementary Figure 4, Panel A), supporting the localization of LRRK2 to endolysosomal membranes that are likely BMP-enriched. Our manuscript's Results and Methods sections have been updated accordingly.

Does the mutant affect endolysosomes?

As referenced in our original manuscript, several studies have reported that pathogenic LRRK2 mutations can lead to endolysosomal defects. Consistent with these reports, we also observed morphological alterations in endolysosomes of cells expressing mutant LRRK2, including reduced MVE size and fewer ILVs, as shown in Figure 1D–F. These observations are in agreement with previously described phenotypes associated with pathogenic LRRK2 variants. Furthermore, in mutant LRRK2 MEFs, and now in humanderived fibroblasts (see new Figure 7, Panel A and B), we observed a decrease in BMP immunostaining signal.

(3) Immunofluorescence data indicate that BMP is decreased in mutant LRRK2expressing cells compared to WT (Figure 1A-B), but mass spec data indicate that di-22:6BMP and di-18:1-BMP are increased (Figure 3). Authors conclude that the BMP pool detected by mass spec in mutant cells is less antibody-accessible than that present in wt cells, or that the anti-BMP antibody is less specific and that it detects other analytes. This is an awkward conclusion, since the IF signal with the antibody is lower (not higher): why would the antibody be less specific? Could it be that the antibody does not see all BMP isoforms equally well? Moreover, the observations that mutant cells contain smaller MVEs (Figure 1D-F) with fewer ILVs are consistent with the IF data and reduced BMP amounts. This needs to be clarified.

As previously reported by us (Lu et al., J Cell Biol 2022;221:e202105060) and others (Berg AL, et al. Cancer Lett. 2023, 557:216090), discrepancies can occur between BMP levels detected by immunofluorescence and those quantified by mass spectrometry. This is because immunostaining reflects the pool of antibody-accessible BMP, whereas lipidomics measures the total cellular content of all BMP molecular species, irrespective of their distribution or accessibility.

We agree that the anti-BMP antibody may not detect all BMP isoforms equally well. Differences in acyl chain composition (such as the degree of saturation or chain length) can alter the stereochemistry of BMP and, consequently, epitope accessibility to antibody binding.

In addition, in a personal communication with Monther Abu-Remaileh (Stanford University), we were informed that the antibody may also cross-react with other lipid species in endolysosomes. Nevertheless, since there is no formal evidence supporting this, we have removed the sentence in the Discussion section stating “Alternatively, the antibody may also detect non-BMP analytes” to avoid any potential misinterpretations. In its place, we have added a short statement noting that “not all BMP isoforms may be detected equally well”.

Mass spectrometry data are only shown for two BMP species (di-22:6, di-18:1). What are the major BMP isoforms in WT cells? The authors should show the complete analysis for all BMP species if they wish to draw quantitative conclusions about the amounts of BMP in wt and mutant cells. Finally, BMP and PG are isobaric lipids. Fragmentation of BMPs or PGs results in characteristic fingerprints, but the presence of each daughter ion is not absolutely specific for either lipid. This should be clarified, e.g., were BMP and PG

separated before mass spec analysis? Was PG affected? The authors should also compare the BMP data with mass spec data obtained with a control lipid, e.g., PC.

Regarding BMP isoforms, our targeted UPLC-MS/MS analyses revealed that 2,2'-di-22:6-BMP (sn2/sn2') and 2,2'-di-18:1-BMP (sn2/sn2') are the predominant BMP isoforms in MEF cells, consistent with previous reports showing docosahexaenoyl (22:6; DHA) and oleoyl (18:1) BMP as the most abundant isoforms. Across diverse mammalian cells and tissues, BMP typically exhibits a fatty acid composition dominated by oleoyl, with polyunsaturated fatty acids (particularly DHA) also contributing substantially. Enrichment of DHA-containing BMP species has been observed in multiple systems, including rat uterine stromal cells, PC12 cells, THP-1 and RAW macrophages, as well as in rat and human liver. This consistent presence of oleoyl- and docosahexaenoyl-containing BMP species across tissues indicates that these acyl chains are conserved features influencing the lipid's structural and functional characteristics (Kobayashi et al. J Biol Chem, 2002; Hullin-Matsuda et al. Prostaglandins Leukotriens Essent Fatty Acids, 2009; Thompson et al. Int J Toxicol. 2012; Delton-Vandenbroucke et al. J Lipid Res, 2019).

Nevertheless, we have included a Table (Panel H in updated Supplemental Figure 1) showing other BMP species that were also detected in our lipidomics analysis. Overall, dioleoyl (18:1)- and di-docosahexaenoyl (22:6)-BMP species were the most abundant in MEF cells, whereas di-arachidonoyl (20:4)- and di-linoleoyl (18:2)-BMP isoforms were present at lower levels. Consistently, R1441G LRRK2 MEFs displayed higher levels of dioleoyl- and di-docosahexaenoyl-BMP compared with WT cells, and these elevations were reduced following LRRK2 kinase inhibition with MLI-2. Data from three independent representative experiments are shown, and the manuscript has been revised accordingly to include these results.

Regarding the separation of BMP and PG species, we confirm that BMP and PG were chromatographically resolved prior to MS/MS detection using a validated UPLC-MS/MS method developed by Nextcea, Inc. PG exhibits a substantially longer LC retention time than BMP, ensuring complete baseline separation. This approach (established by Nextcea nearly two decades ago and later validated through a multi-year collaboration with the U.S. FDA to clinically qualify di-22:6-BMP as a biomarker) prevents any ambiguity arising from the isobaric nature of BMP and PG species. No changes in PG levels were detected under any experimental conditions.

Finally, we employed isotope-labeled BMP as an internal standard to ensure robust normalization across samples. These additional details and references cited above have been included in the revised Methods and References sections to further clarify the analytical rigor of our lipidomics workflow.

(4) It is quite surprising that the amounts of labeled BMP continue to increase for up to 24h after a short 25min pulse with heavy BMP precursors (Figure 4B).

In these isotope-labeling experiments, it is important to note (as described in our original manuscript) that two distinct pools of metabolically labeled BMP species were detected: semi-labeled BMP (with only one heavy isotope-labeled fatty acyl chain) and fully-labeled BMP (with both fatty acyl chains labeled). We consider the fully-labeled BMP pool to provide the most reliable readout for BMP turnover, as it showed a rapid decline after a 1h chase (decreasing by more than 50% within 8 h in all conditions), reaching its lowest levels at the end of the 48-h chase period.

The apparent increase in semi-labeled BMP species over time may be explained by continued incorporation of labeled precursors following the initial pulse. Specifically, once existing semi-labeled and fully-labeled BMP molecules are degraded by PLA2G15 (Nyame K, et al. Nature 2025, 642:474-483), the resulting isotope-labeled lysophosphatidylglycerol (LPG) and

fatty acids could be recycled and re-enter a new round of BMP biosynthesis, leading to a gradual accumulation of semi-labeled BMP such as di-18:1-BMP. Why would this reasoning not also apply to the fully-labeled species? Once the pulse is completed, newly incorporated non-labeled fatty acyl chains present in the cellular pool can compete with labeled ones during subsequent rounds of lipid remodeling or synthesis. As a result, the probability of generating semi-labeled BMP molecules becomes higher than that of forming fully-labeled species. Consistent with this, our data show an increase in only semi-labeled BMP species (but not in fully-labeled ones) up to 24 hours after the pulse. We have added a clarification regarding this point in the revised manuscript.

(5) It is argued that upregulation of CLN5 may be due to an overall upregulation of lysosomal enzymes, as LAMP2 levels were also increased (Figure 2A, C, E). Again, this is not consistent with the observed decrease in MVE size and number (Figure 1D-F). As mentioned above, other independent markers of endo-lysosomes should be analyzed (eg, LAMP1, CD63, RAB7), and/or other lysosomal enzymes (e.g. cathepsin. D).

Our revised manuscript now includes new immunofluorescence data for BMP, LAMP1 and CD63 (New Figure 7, Panels A-F) together with biochemical analysis of CD63 protein levels (New Supplemental Figure 4, Panel B) in human skin fibroblasts derived from healthy controls and LRRK2 G2019S PD patients. Quantitative analysis of these experiments revealed no statistically significant differences in total cellular levels of either LAMP1 or CD63 between groups. However, our results consistently show increased CLN5 protein levels in both mouse and human fibroblast cell lines harboring pathogenic LRRK2 mutations. Upregulation of CLN5 may reflect a compensatory effect from loss of BMP via EV exocytosis. As discussed above, the elevated LAMP2 signal observed in the engineered MEF clone expressing R1441G could represent a cell type-specific effect, potentially linked to differential penetrance of LRRK2 signaling on the lysosomal biogenesis response. Our Results and Discussion sections have been updated accordingly.

(6) The authors report that the increase in BMP is not due to an increase in BMP synthesis (Figure 4), although they observe a significant increase in CLN5 (Figure 5A) in LRRK2 mutant cells. Some clarification is needed.

In our original manuscript, we proposed that although CLN5 protein levels are increased in R1441G LRRK2 MEFs, the absence of significant changes in BMP synthesis rates (Figure 4B, C) may reflect either limited substrate availability or that CLN5 is already operating near its maximal enzymatic capacity. Our new subcellular fractionation data (new Figure 7, Panel A) further indicate that, despite a relative increase in total CLN5 levels in G2019S LRRK2 human fibroblasts, the amount of CLN5 associated with endolysosomes remains comparable between mutant LRRK2 and control cells. This suggests that a considerable fraction of upregulated CLN5 may not localize to endolysosomes, potentially accumulating in the endoplasmic reticulum due to enhanced translation or impaired trafficking. Unfortunately, the available anti-CLN5 antibody did not yield reliable immunofluorescence signals, preventing us from directly confirming this possibility. Nevertheless, in light of our new data (new Supplemental Figure 4A), we have included a clarification in the revised manuscript discussing this possibility as well.

(7) Authors observe that both LAMP2 and BMP are decreased in EVs by GW4869 and increased by bafilomycin (Figure 6). Given my comments above on Figure 1, it would also be nice to illustrate/quantify the effects of these compounds on cells by immunofluorescence.

We appreciate the reviewer's suggestion. We have previously published immunofluorescence data showing increased BMP accumulation in endolysosomes following treatment with bafilomycin A1 Lu A, et al. J Cell Biol. 2009, 184:863-879). However, in the present study, our lipidomics analyses revealed a decrease in both di22:6-BMP and di-18:1-BMP species in cells

treated with this compound. As discussed above, this apparent discrepancy likely reflects methodological differences between immunofluorescence, which detects only antibody-accessible BMP pools, and lipidomics, which quantifies total cellular BMP content.

Moreover, in a recent study (Andreu Z, et al. Nanotheranostics 2023, 7:1-21), BMP levels were analyzed by immunofluorescence in cells treated with spiroepoxide, a potent and selective irreversible inhibitor of nSMase (different from GW4869) known to block EV release. Spiroepoxide-treated cells showed decreased BMP immunostaining; a result that, again, does not align with mass spectrometry data revealing increased cellular BMP levels upon GW4869 treatment. Notably, in that study, spiroepoxide was used instead of GW4869 because the intrinsic autofluorescence of GW4869 could potentially interfere with the immunofluorescence BMP signal.

We therefore consider lipidomics measurements to provide a more reliable and quantitative representation of BMP dynamics under these conditions.

Reviewer #1 (Recommendations for the authors):

Major concerns:

(1) 48 h for MLI2 treatment seems too long. LRRK2 kinase activity is inhibited with much shorter incubation times. The longer the incubation, the more likely off-target effects are. The authors should repeat these experiments with 1-2 h of MLI2.

We thank the reviewer for this valuable comment. We acknowledge that MLI-2 is a potent and selective LRRK2 kinase inhibitor that achieves near-complete target engagement within a few hours of treatment. However, prolonged exposure has been widely used in the field without evidence of significant off-target effects. Several studies, including Fell et al. (2015, J Pharmacol Exp Ther 355:397-409), De Wit et al. (2019, Mol Neurobiol 56:5273-5286), Ho et al. (2022, NPJ Parkinson's Dis 8:115), Tengberg et al. (2024, Neurobiol Dis 202:106728), and Jaimon et al. (2025, Sci Signal 18:eads5761), have employed long-term (24-48 h) MLI-2 treatments at comparable concentrations without detecting toxicity or off-target alterations, including in MEFs (Ho et al., 2022; Dhekne et al., 2018, eLife 7:e40202).

In our study, 48-hour incubations were necessary to sustain full LRRK2 inhibition throughout the extracellular vesicle (EV) collection period. EV biogenesis, BMP biosynthesis, and packaging into EVs are time-dependent processes; therefore, extended incubation and collection periods (48 h) were required to allow downstream effects of LRRK2 inhibition on BMP production and release to manifest, and to obtain sufficient EV material for biochemical and lipidomic analyses. This experimental design also reflects our and others' previous observations in humans and non-human primates, where urinary BMP changes are associated with chronic or subchronic LRRK2 inhibitor treatment (Baptista MAS, Merchant K, et al. Sci Transl Med. 2020, 12:eaav0820; Jennings D, et al. Sci Transl Med. 2022, 14:eabj2658; Maloney MT, et al. Mol Neurodegener. 2025, 20:89). Importantly, under these conditions, we did not observe significant changes in cell viability or morphology, supporting that the treatment was well tolerated.

We have clarified this rationale in the revised Methods section to emphasize that the prolonged incubation reflects the experimental design for EV isolation rather than a requirement for achieving LRRK2 inhibition.

(2) Is there a reason why the authors don't include CD81, CD63, and Syntenin-1 in their study as an EV marker? Using solely Flotillin-1 does not seem to be enough to justify their claims.

We actually used not only Flotillin-1 but also LAMP2 as EV markers in our study. While both Flotillin-1 and LAMP2 detection on EVs may vary depending on the cell type, we and others

have reported enrichment of Flotillin-1 and LAMP proteins in isolated small EV fractions (Kowal et al., 2016; Lu et al., 2018; Mathieu et al., 2021; Ferreira et al., 2022). In particular, one of these studies reported that “LAMP1-positive subpopulations of EVs represent MVB/lysosome-derived exosomes, which also contain syntenin-1.” Therefore, our choice of EV markers (LAMP2 and Flotillin-1) is consistent with those previously and reliably used to characterize small EVs.

Nevertheless, to further address the reviewer’s concern, we performed additional experiments using a CD63-based fluorescence sensor (CD63-pHluorin), which, combined with TIRF microscopy, enables real-time visualization of CD63-positive exosome release. These experiments were conducted in control and LRRK2-mutant fibroblasts, and the data are presented in new Figure 7 (Panels G-I; Videos 1 and 2). We have also included all relevant references and clarified this point in the revised manuscript.

(3) Indeed, to quantify the amount of certain proteins in EVs, the authors should normalize them by CD63 or CD81.

Protein normalization in isolated EV fractions is indeed challenging. Although tetraspanins such as CD63 and CD81 are commonly enriched in EVs, their abundance can vary considerably across EV subpopulations, cell types, and experimental conditions, making them unreliable as universal normalization markers (Théry et al., J Extracell Vesicles, 2018; Margolis & Sadovsky, Nat Rev Mol Cell Biol, 2019). Current guidelines from the International Society for Extracellular Vesicles (ISEV), as described in the Minimal Information for Studies of Extracellular Vesicles 2018 (MISEV2018; Théry C, et al. J Extracell Vesicles. 2018, 7:1535750) and updated in MISEV2024 (Welsh JA, et al. J Extracell Vesicles. 2024, 13:e12404), recommend reporting multiple EV markers rather than relying on a single protein for normalization. They also suggest ensuring comparable experimental conditions by using the same number of cells at the start of the experiment and normalizing EV data to cell number or whole-cell lysate protein content at the end of the experiment, among other approaches.

In our study, we normalized EV data to whole-cell lysate (WCL) protein content, as this approach accounts for differences in EV production due to variations in cell number or treatment conditions and is commonly used in the field (Kowal et al., PNAS, 2016; Mathieu et al., Nat Commun, 2021). We also included Flotillin-1 and LAMP2 as EV markers, both of which have been validated as molecular markers of small EV subpopulations.

(4) Hyper normalization in WB quantification in Figure 2E-G is statistically incorrect, as it assumes that one group (in this case, R1441G ctrl) has no variability at all, which is not biologically possible. The authors should repeat the quantification without hypernormalizing one of their groups. This issue is prevalent across the whole manuscript.

We understand the concern regarding “hyper-normalization” (i.e., expressing all values relative to one condition set to 1), which may mask variability in the reference group. However, it is standard practice in immunoblotting analysis to express data relative to a control condition for comparison, as variations in membrane transfer, exposure time, and signal development can differ across blots. In our case, the data are expressed as relative levels (arbitrary units) rather than absolute quantitative values. To facilitate comparison between datasets and account for inter-experimental variation, we continued to express values relative to the mutant LRRK2 MEF condition.

On the other hand, in lipidomics experiments, despite using the same number of seeded cells and identical extraction and analysis protocols, minor biological and technical variability was observed across independent replicates. This variability is inherent to the experimental system and is now explicitly represented in the new table included in Supplemental Figure

1F, which compiles three independent representative lipidomics experiments showing quantitative BMP levels across different conditions.

(5) The authors perform a t-test in Figure 2E-G when comparing more than 2 groups, which is wrong. The authors should use a two-way ANOVA as they are comparing genotype and treatment.

We appreciate the reviewer's comment and agree with this observation. The MLI-2 and CBE experiments were performed independently and in separate experimental runs; therefore, we have reanalyzed these datasets separately rather than combining them in a two-way ANOVA. To properly compare more than two groups within each dataset, we have now applied a Kruskal-Wallis test followed by an uncorrected Dunn's post hoc test (Figure 2 D-F and H-J). This non-parametric approach is more appropriate for our data structure, as EV experiments are usually subject to high variability and immunoblot quantifications involving small sample sizes ($n \approx 6$) do not always meet the assumptions of normality or equal variance. The Kruskal-Wallis test does not assume normality or equal variances, making it more robust for small, variable biological datasets. The statistical analyses and figure legend have been updated in the revised manuscript accordingly.

In addition, since our CBE treatments yielded statistically non-significant data, we have softened our conclusions throughout the manuscript concerning the contribution of GCase activity to EV-mediated BMP release modulation.

(6) There is a very strong reduction in flotillin-1 in R1441G cells vs WT (Figure 2G) in the EV fraction. That reduction is further exacerbated with MLI2, which likely means it is not kinase activity dependent. Can the authors comment on that?

We agree with the reviewer that Flotillin-1 showed a different behavior compared with LAMP2 in these experiments. As recommended by the MISEV guidelines (Théry C, et al. J Extracell Vesicles. 2018; 7:1535750; Welsh JA, et al. J Extracell Vesicles. 2024, 13:e12404), it is important to analyze more than one EV-associated protein marker. We examined LAMP2, which, together with LAMP1, has been reported to be specifically enriched in EVs of endolysosomal origin (exosomes; Mathieu et al., Nat Commun. 2021, 12:4389). In contrast, Flotillin-1 is also associated with small EVs but may represent a distinct EV subpopulation from those positive for LAMP proteins (Kowal J, et al. PNAS 2016, 113:E968-E977).

Nevertheless, the biochemical analysis of isolated EV fractions was complemented by our lipidomics data and, in the revised version, by TIRF microscopy analysis of exosome release in control and G2019S LRRK2 human fibroblasts (new Figure 7, Panels G-I; Videos 1 and 2). In this analysis, we confirmed increased exocytosis of CD63-pHluorin– positive endolysosomes in G2019S LRRK2 human fibroblasts compared to controls, an effect that was reversed by MLI-2 treatment. The CD63-pHluorin–positive compartment of these cells was also largely positive for BMP (new Figure 7G). Collectively, these findings further support the regulatory role of LRRK2 activity in EV-mediated BMP secretion.

(7) In Figure 2C, the authors should express that the LAMP2-EV and flotillin-1 EV fractions from the WB are highly exposed. As presently presented, it is slightly misleading.

We thank the reviewer for this comment. In EV preparations, the amount of protein recovered is typically very low. Therefore, although we loaded all the EV protein obtained from each sample, the immunoblots for LAMP2 and Flotillin-1 in EV fractions required longer exposure times to visualize clear signals across all conditions. We have now indicated in the corresponding figure legend that these EV blots are long-exposure blots to facilitate signal detection and avoid any potential misunderstanding.

(8) If Figure 2C and D are from two different experiments, they should not be plotted together in Figure 2E-G. You cannot compare the effect of MLI2 vs CBE if done in

completely different experiments.

We appreciate the reviewer's comment and agree with this observation. The MLI-2 and CBE experiments were performed independently and in separate experimental runs; therefore, we have reanalyzed these datasets separately rather than combining them in a two-way ANOVA. To properly compare more than two groups within each dataset, we have now applied a Kruskal-Wallis test followed by an uncorrected Dunn's post hoc test (Figure 2 D-F and H-J). This non-parametric approach is more appropriate for our data structure, as EV experiments are usually subject to high variability and immunoblot quantifications involving small sample sizes ($n \approx 6$) do not always meet the assumptions of normality or equal variance. The Kruskal-Wallis test does not assume normality or equal variances, making it more robust for small, variable biological datasets. The revised statistical analyses and figure legends have been updated accordingly in the manuscript.

(9) *The authors state that "For the R1441G MEF cells, MLI-2 decreased EV concentration while CBE increased EV particles per ml, in agreement with the effects observed in our biochemical analysis." As Figure S1D shows no statistical significance, the authors don't have sufficient evidence to make this claim.*

We apologize for this overstatement. We have revised the text to clarify that, although the differences did not reach statistical significance, a consistent trend toward decreased EV concentration upon MLI-2 treatment and increased EV release following CBE treatment was observed in R1441G MEF cells.

(10) *"Altogether, given that BMP is specifically enriched in ILVs (which become exosomes upon release), the data presented above support our biochemical analysis (Figure 2C, D, F) and suggest a role for LRRK2 and GCase in modulating BMP release in association with LAMP2-positive exosomes from MEF cells." As Figure 3E shows no statistical difference of BMP on EVs upon CBE treatment, this sentence is not accurate and should be reframed. Furthermore, the authors claim an increase in EV-LAMP2 in R1441G cells compared to WT, however, the amount of BMP in EVs of R1441G cells vs WT is unchanged with a non-significant reduction. This contradiction does not support the authors' conclusions and really puts into question their whole model.*

We thank the reviewer for this observation. After reanalyzing our biochemical data from isolated EV fractions (see new Panels D-F and H-J) using an improved statistical approach, we found that although EV-associated LAMP2 levels were consistently elevated in untreated R1441G LRRK2 MEFs compared to WT cells, CBE treatment only produced a non-significant trend toward increased EV-associated LAMP2 compared to untreated R1441G LRRK2 cells. Accordingly, we have revised the sentence to read as follows:

"Altogether, given that BMP is specifically enriched in ILVs (which become exosomes upon release), the data presented above support our biochemical analysis (Figure 2C, E, G, I) and suggest that LRRK2 activity regulates BMP release in association with LAMP2positive exosomes, whereas GCase activity appears to have a more variable effect under the tested conditions."

We also agree with the reviewer that, in our MEF model, the amount of BMP in EVs of R1441G cells vs WT is unchanged with a non-significant reduction. However, pharmacological modulation supports our conclusion that BMP release is modulated by LRRK2 activity. Specifically, treatment with the LRRK2 inhibitor MLI-2 decreased EV-associated BMP and LAMP2 levels in R1441G LRRK2 MEFs, and our new data (new Figure 7, Panel G-I; Videos 1 and 2) show increased exocytosis of CD63-pHluorin- positive endolysosomes in G2019S LRRK2 human fibroblasts compared to controls, an effect that was reversed by MLI-2 treatment. The CD63-pHluorin-positive compartment of these cells was also largely positive for BMP (new Figure 7G).

In light of the reviewer's comment about CBE treatment, we have softened our conclusions throughout the manuscript concerning the contribution of GCase activity in this model.

(11) *In Figure 5, 16 h of MLI2 treatment is too long and can lead to off-target effects. I would advise reducing it to 1-4 h.*

Prolonged MLI-2 treatments have been extensively used in the field without evidence of significant off-target effects. Several studies, including Fell et al. (2015, J Pharmacol Exp Ther 355:397-409), De Wit et al. (2019, Mol Neurobiol 56:5273-5286), Ho et al. (2022, NPJ Parkinson's Dis 8:115), Tengberg et al. (2024, Neurobiol Dis 202:106728), and Jaimon et al. (2025, Sci Signal 18:eads5761), have applied long-term (24-48 h) MLI-2 treatments at comparable concentrations without detecting toxicity or off-target alterations, including in MEFs (Ho et al., 2022; Dhekne et al., 2018, eLife 7:e40202). Moreover, the data presented in Figure 5 demonstrate a reduction in CLN5 protein levels in both MEFs and human fibroblasts following MLI-2 treatment, confirming the specificity of the observed effects in LRRK2 mutant cells.

(12) *"Our data suggest that BMP is exocytosed in association with EVs and that LRRK2 and GCase activities modulate BMP secretion." Again, cells carrying the R1441G mutation have the same amount of BMP in EVs than WT. This sentence is not factually accurate. Accordingly, CBE did not change the amount of BMP in EVs.*

We thank the reviewer for this observation and agree that, in our MEF model, the amount of BMP in EVs from R1441G LRRK2 cells is comparable to that observed in WT cells. However, pharmacological modulation supports our conclusion that BMP release is modulated by LRRK2 activity. Specifically, treatment with the LRRK2 inhibitor MLI-2 decreased EV-associated BMP levels in R1441G LRRK2 MEFs, and our new data (new Figure 7G-I; Videos 1 and 2) show increased exocytosis of CD63-pHluorin-positive endolysosomes in G2019S LRRK2 human fibroblasts compared to controls, an effect that was reversed by MLI-2 treatment. The CD63-pHluorin-positive compartment of these cells was also largely positive for BMP (new Figure 7G). These findings further support the regulatory role of LRRK2 activity in EV-mediated BMP secretion. In addition, in light of the reviewer's comment about CBE treatment, we have softened our conclusions throughout the paper concerning the contribution of GCase activity in this model.

(13) *Figure 6; EV release should have been monitored by more accurate markers such as CD63 and CD81.*

We thank the reviewer for this comment. We and others (Kowal et al., 2016; Lu et al., 2018; Mathieu et al., 2021; Ferreira et al., 2022) have reported enrichment of Flotillin-1 and LAMP proteins in isolated small EV fractions. In particular, one of these studies (Mathieu et al., Nat Commun. 2021), in which bafilomycin A1 was also used (to boost exosome release), reported that "LAMP1-positive subpopulations of EVs represent MVB/lysosome-derived exosomes, which also contain syntenin-1." Altogether, our choice of EV markers (LAMP2 and Flotillin-1) is consistent with those previously and accurately used to characterize EVs. We have now included all relevant references in the revised manuscript to further clarify this point.

(14) *Figure 6 suggests that exosomal BMP is controlled by EV release. I would think that is rather obvious.*

We agree that the finding that exosomal BMP release is influenced by EV secretion may appear "obvious." However, our intention in Figure 6 was to provide direct experimental evidence confirming this relationship using pharmacological modulators of EV release. Specifically, inhibition of EV secretion with GW4869 reduced exosomal BMP levels, whereas stimulation with bafilomycin A1 increased them. These data were important to establish a causal link between EV trafficking and BMP export, thereby validating our model and

supporting the interpretation that LRRK2 regulates BMP homeostasis through EV-mediated exocytosis, which is further modulated, to some extent, by GCase activity.

Minor concerns:

(1) *Figure 1: Change colors to be color blind friendly.*

We thank the reviewer for this helpful suggestion. We have adjusted the colors in Figure 1 to be color-blind friendly. In addition, we have applied the same color-blind friendly palette to the new immunofluorescence data presented in new Figure 7, Panel A and D.

(2) *More consistency on "Xmin" vs "X min" would be appreciated.*

We thank the reviewer for this observation. We have revised the manuscript to ensure consistent formatting of time indications throughout the text and figures, using the standardized format "X min."

Reviewer #2 (Recommendations for the authors):

(1) *Figure 2C-D. Were equal amounts of protein loaded in each lane?*

Equal protein amounts were loaded in lanes corresponding to whole-cell lysate (WCL) fractions and normalized based on α -Tubulin levels.

For the extracellular vesicle (EV) fractions, all protein recovered from EV pellets after isolation was loaded. In all EV-related experiments, we seeded the same number of EV-producing cells per condition, and the resulting EV-derived data (from both immunoblotting and lipidomics analyses) were normalized to the corresponding whole cell lysate (WCL) protein content to ensure comparability across conditions.

All these technical details have been included in the Materials section of our revised manuscript.

(2) *The authors refer to the papers of Medoh et al (ref 43) and Singh et al. (44) for the key role of CLN5 in the BMP biosynthetic pathway. However, Medoh et al reported that CLN5 is the lysosomal BMP synthase. In contrast, Singh et al. reported that PLD3 and PLD4 mediate the synthesis of SS-BMP, and did not find any role for CLN5.*

To avoid any confusion or misinterpretation of our findings regarding CLN5 and given that we do not analyze PLD3 or PLD4 in our study, we have decided to replace the reference to Singh et al. with Bulfon D. et al. (Nat. Commun. 2024, 15:9937) instead. This last work, conducted by an independent group distinct from the one that originally described CLN5, also validated CLN5 as the sole BMP synthase in cells.

Also, authors mention that bafilomycin A1 (B-A1) dramatically boosts EV exocytosis, referring to Kowal et al., 2016 (ref 35) and Lu et al., 2018 (ref 45). However, this is not shown in Kowal et al.

We thank the reviewer for pointing out this mistake. We apologize for the incorrect citation and have now corrected the reference. The statement regarding the effect of bafilomycin A1 on EV exocytosis now appropriately refers to Mathieu et al., 2021 and Lu et al., 2018.

(3) *Page 7, it is stated that "No statistically significant differences in intracellular BMP levels were observed in WT LRRK2 MEFs upon LRRK2 or GCase inhibition(Supplemental Figure 1D, E)". The authors probably mean "Supplemental Figure 1F, G"*

We thank the reviewer for noting this error. We have corrected the text to refer to panels F and G of Supplemental Figure 1, which correspond to the relevant data. We have also revised the reference to panel I of Supplemental Figure 1 accordingly.

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